



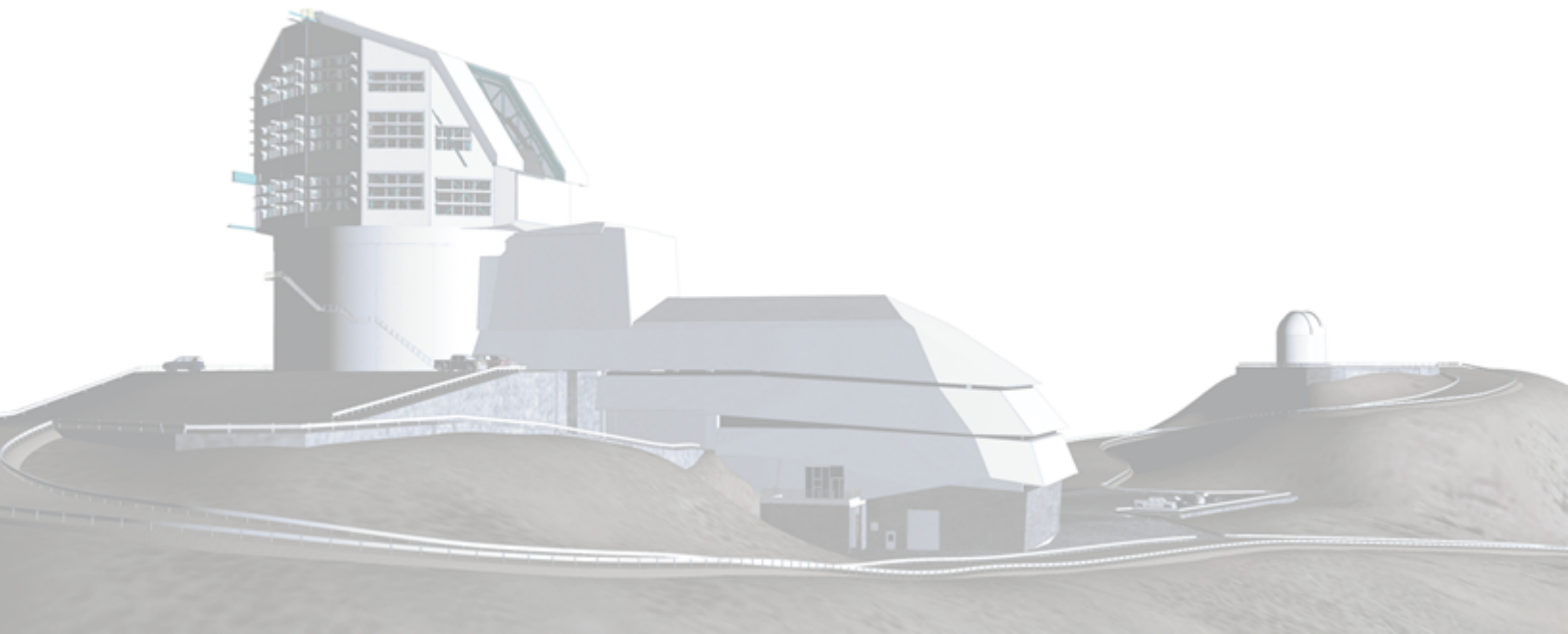
Vera C. Rubin Observatory
Data Management

AIDA User Manual

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rel1.1

Latest Revision: 2026-10-02



Abstract

This document is the software user manual of the web application AIDA (Advanced Infrastructure for Data Analysis), tool provided as deliverable for the In-Kind Contribution ITA-INA S23: *“Staff effort in support of Rubin commissioning: ML tools for instrumental monitoring and analysis”*.

AIDA is a portable and modular web application, designed to provide an efficient and intuitive software infrastructure to support monitoring of data acquiring systems over time, diagnostics and both scientific and engineering data quality analysis, particularly suited for astronomical instruments. Its versatility makes it possible to extend its functionalities, by integrating and customizing monitoring and diagnostics systems, as well as scientific data analysis solutions, including machine learning and data mining techniques and methods.

Change Record

Version	Date	Description	Owner name
1.0	2024-05-01	Initial release.	Giuseppe Riccio
1.1	2026-09-17	Manual of AIDA version 1.0, including all functionalities foreseen in the in-kind contribution workplan. Improved ML section.	Giuseppe Riccio

Document curator: Giuseppe Riccio

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AIDA User Manual

1 Introduction

This document is the software user manual of the web application AIDA (Advanced Infrastructure for Data Analysis). AIDA is a portable and modular web application, designed to provide an efficient and intuitive software infrastructure to support monitoring of data acquiring systems over time, diagnostics and both scientific and engineering data quality analysis, particularly suited for astronomical instruments.

The AIDA design was mainly focused on the following goals:

- Daily mission operations, for instruments health assessment and reporting:
 - Monitoring and verifying the nominal quality of data;
 - Production of user-defined reports (automatic and/or on-demand);
- Short/Mid/Long term activities, to investigate and report instruments behaviour over time:
 - Analysis of instruments HouseKeeping/TeleMetry (HK/TM) trends;
 - Analysis of instruments systematic effects;
 - Detection and analysis of instrumental features, degradation and anomalies;
- Data Analysis:
 - Basic and on-demand advanced statistics for data correlation and quality assessment;
 - Machine Learning based classification/regression/clustering on data.

In the following we will use the term AidaHM, referring to a generic hosting machine on which AIDA is installed and running.

Main AIDA functionalities are:

- **Instrument monitoring, report generation and delivery**

- periodic report generation on a user-defined parameters list;
- customized on demand report generation on a user selected parameter list;

- **Visualization/Exploration**

- Series of plots on user selected parameters/data products and ranges;
- Observed images (static view, dynamic windowing, statistical characterization);

- **Statistics**

- Standard (default) estimators;
- Special estimations:
 - mode, percentiles, biweight, σ -clipping; skewness;
- Machine Learning based classification/regression/clustering on local data.

- **Additional features:**

- Flagging System: association of a semaphore-like flag to each kind of experiment, to indicate its status and generate a related PDF report, summarizing its results and flags;
- Local Data Analysis: generation of plots on data and visualize/analyze images uploaded from local machine;
- Users and System Management:
 - online user registration to be confirmed by administrators;
 - User password recovery system by administrators;
 - enable, disable or remove users;
 - set Operating Mode (Nominal, Commissioning, Contingency);
 - enable/disable and configure systems to monitor;
 - set web app configuration (SMTP server, number of processors to use...);
- Logging System: each operation performed is logged into local DB;
- Easy Step-by-step Installation Procedure;
- Customizable plots graphics;
- Data backup system

In the official version of AIDA, two sources of data are available:

- Rubin Observatory Telemetry data from the Engineering and Facility Database (EFD)¹;
- Fake data of temperature, voltage and current of three generic subsystems in a limited date range ([2023-01-01, 2023-02-12]), stored on a local DB, automatically installed during AIDA installation.

¹<https://obs-controls.lsst.io/Control-User-Interfaces/EFD.html>

2 Installation Procedure

The installation of the AIDA package on the AidaHM is automated by exploiting the docker virtualization technology. AIDA is developed as a double container, i.e. two standardized software units, respectively, the web application and the local database, allowing to isolate the application environment from the rest of the hosting machine. Figure 1 shows the architecture of the docking system.

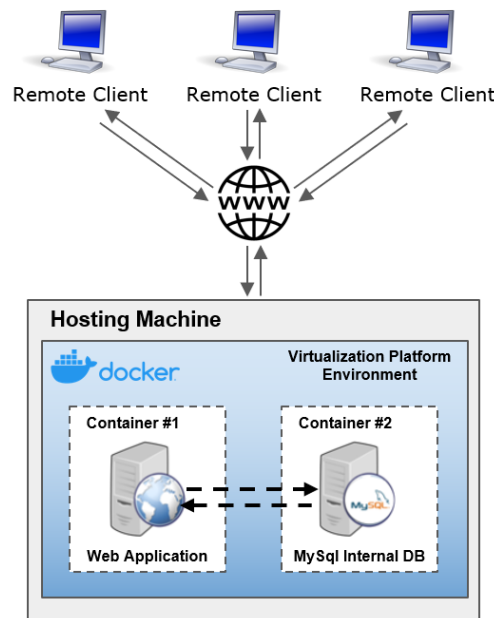


Figure 1: AIDA application environment installed through a docking system.

NOTE: In AIDA , an additional Docker container is available, containing a set of fake data for demonstration purposes². It will be automatically installed during the installation procedure described in Sec.2.1.

2.1 Installation

Installation procedure:

1. Install Docker³ and Docker-Compose⁴. For instance, using Linux Ubuntu, it can be done

²Fake data are available in date range [2023-01-01, 2023-02-12]

³<https://docs.docker.com/engine/install/>

⁴<https://docs.docker.com/compose/install/>

using the following command line:

a. `sudo apt install docker docker-compose -y`

2. Download the complete AIDA files package⁵:

- a. **docker-compose.yml**, the file defining the inner relations of the docker containers;
- b. **Dockerfile**, the docker configuration;
- c. **aidaCodeXX.tar.gz**, containing the aida code;
- d. **apache2.conf**, needed for setting the environment of apache in the docker container (the file should be in the same folder of Dockerfile and could not be deleted);
- e. **php.ini**, needed for setting the environment of php in the docker container (the file should be in the same folder of Dockerfile and could not be deleted);
- f. **aidaDBXX.tar.gz**, containing an empty pre-configured DB.
- g. **AidaDBFakeXX.tar.gz**, containing a set of fake data for demonstration purposes.

3. Uncompress the DBs and code files:

- a. The name of the three directories (two for DBs and one for code, respectively) can be arbitrarily set. But it is important to make the two AIDA directories case sensitive (not required for the fake data DB). This is particularly required in case of an OS not case sensitive (such as MS Windows). To do that, there is a specific command line. For instance, in MS Windows, by running a powershell as administrator:

- `fsutil.exe file SetCaseSensitiveInfo <YourLocalFolder>/aida enable`
- `fsutil.exe file SetCaseSensitiveInfo <YourLocalFolder>/mysqlData enable`

- b. In Linux you need to uncompress the folder as root to correctly preserve the ownership of the code directory (`sudo tar -xvzf aidaCodeXXtar.gz`).

4. Specify the location of the folders (DBs and code) in docker-compose.yml (see Appendix A);
5. Change, if needed, the exposed ports for AIDA (web application and SQL) in docker-compose.yml. This is required only if you already are exposing different services on the default ports (80 for AIDA and 3306 for SQL, see appendix in Appendix A);
6. Launch docker-compose using your docker-compose.yml. This could depend on the OS. On Ubuntu: from the folder where you downloaded docker-compose.yml **AND** Dockerfile launch: `docker-compose up -d -build`

⁵<https://github.com/pepric/aida/tree/main/installation>

NOTE: in <root>/dockerinstall directory, the latest version of Dockerfile (see Appendix B), php.ini and apache2.conf files are located. Moreover the `docker-compose-generator.py` script is provided, to automatically generate the `docker-compose.yml` file

2.2 First access configuration

Once installed, AIDA can be accessed by any web browser at the dedicated url. On first access, an introductory page is shown in order to start the definition of some primary settings (Figure 2):

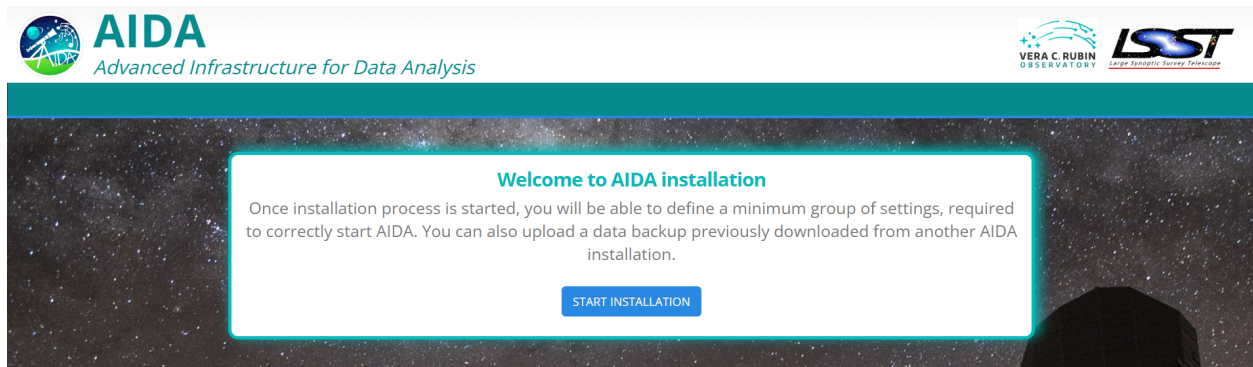


Figure 2: AIDA landing page at first access.

By clicking on “START INSTALLATION” button, a multi-step configuration wizard starts.

2.2.1 Backup Import

At this stage, the user can import a backup previously created on a previous AIDA instance (Figure 3). This optional step can be skipped by clicking on the “Skip” button. To import an existing backup (as a tar.gz archive), the user can easily browse the local machine and upload it. Once uploaded, the user can select the sections to import (Figure 4). If the option “SMTP settings” is not checked, then the AIDA SMTP server should be configured (step 2). If the option “Users” is not checked, then the first AIDA administrator should be registered (step 3). If this step is skipped, then both SMTP server and first administrator should be defined in next steps.

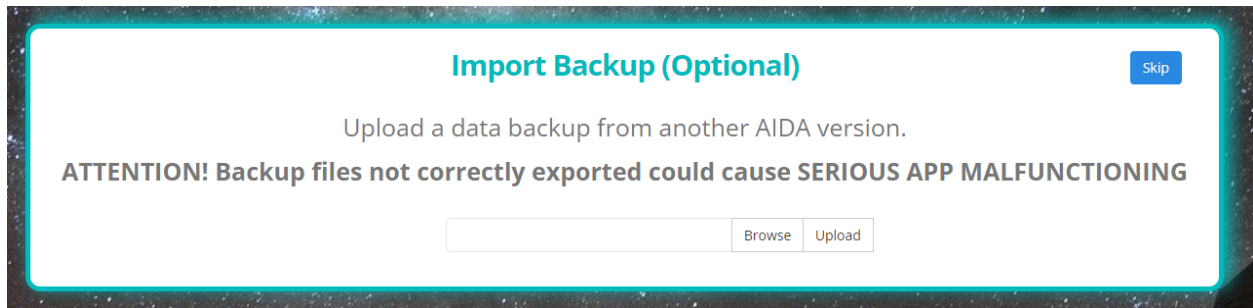


Figure 3: First step of backup import stage of AIDA installation.

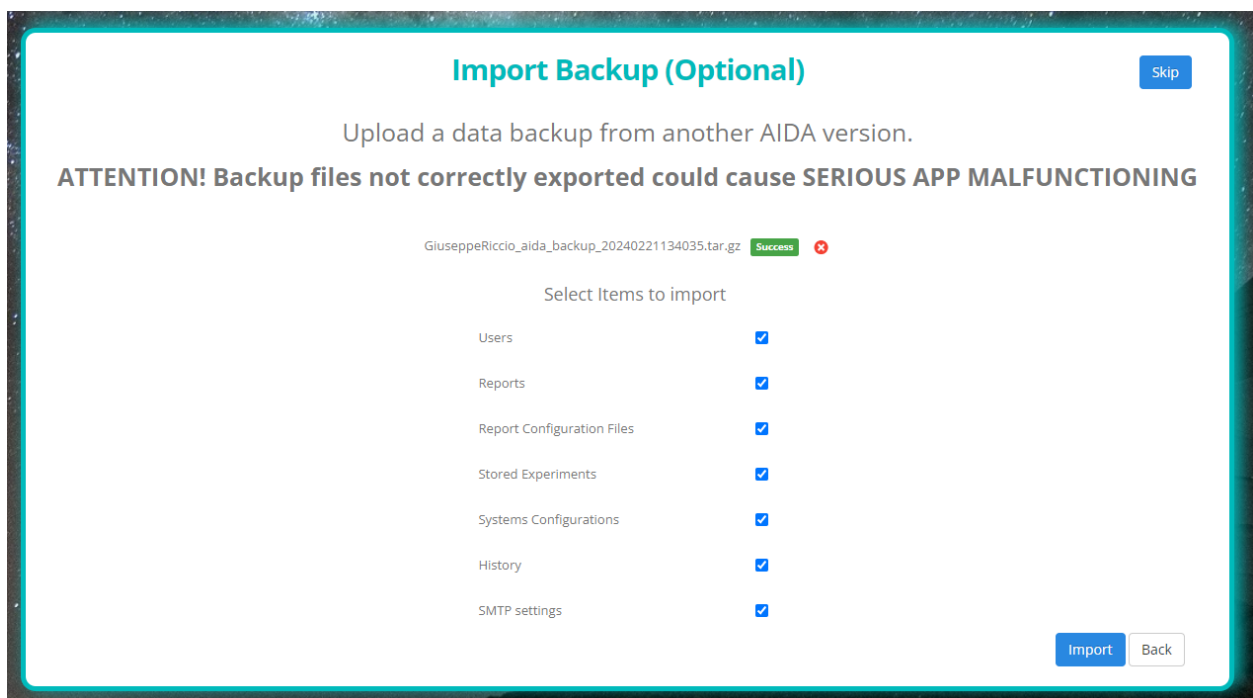
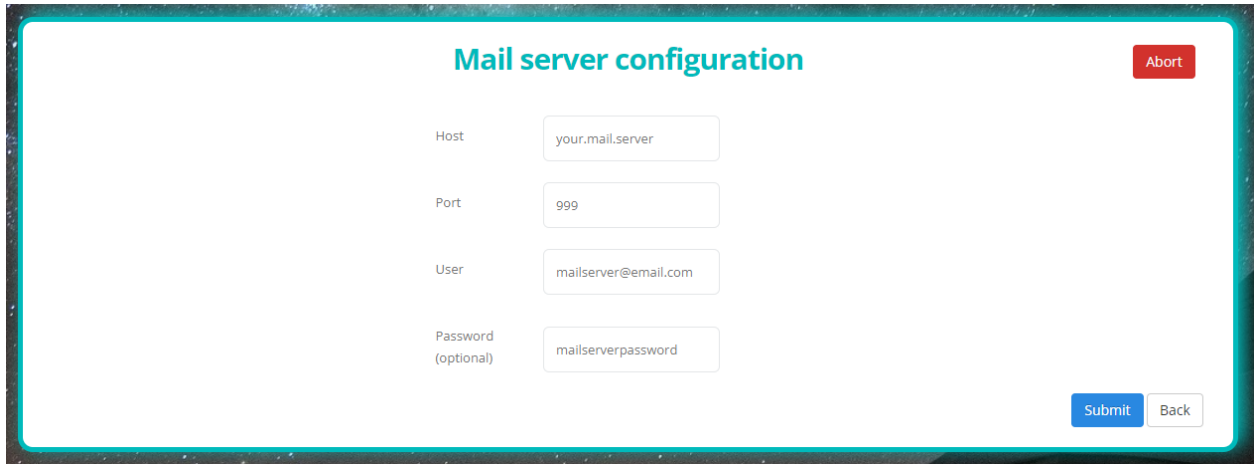


Figure 4: Second step of backup import stage of AIDA installation.

2.2.2 Mail Server Configuration

In order to send e-mails to the users (communications and analysis results), AIDA requires access to a mail server. If not already imported, SMTP server data must be defined by filling this form (Figure 5) with server name, port, username and, if needed, the password.

NOTE: DO NOT USE Google SMTP because of its policy on external apps.



Mail server configuration Abort

Host

Port

User

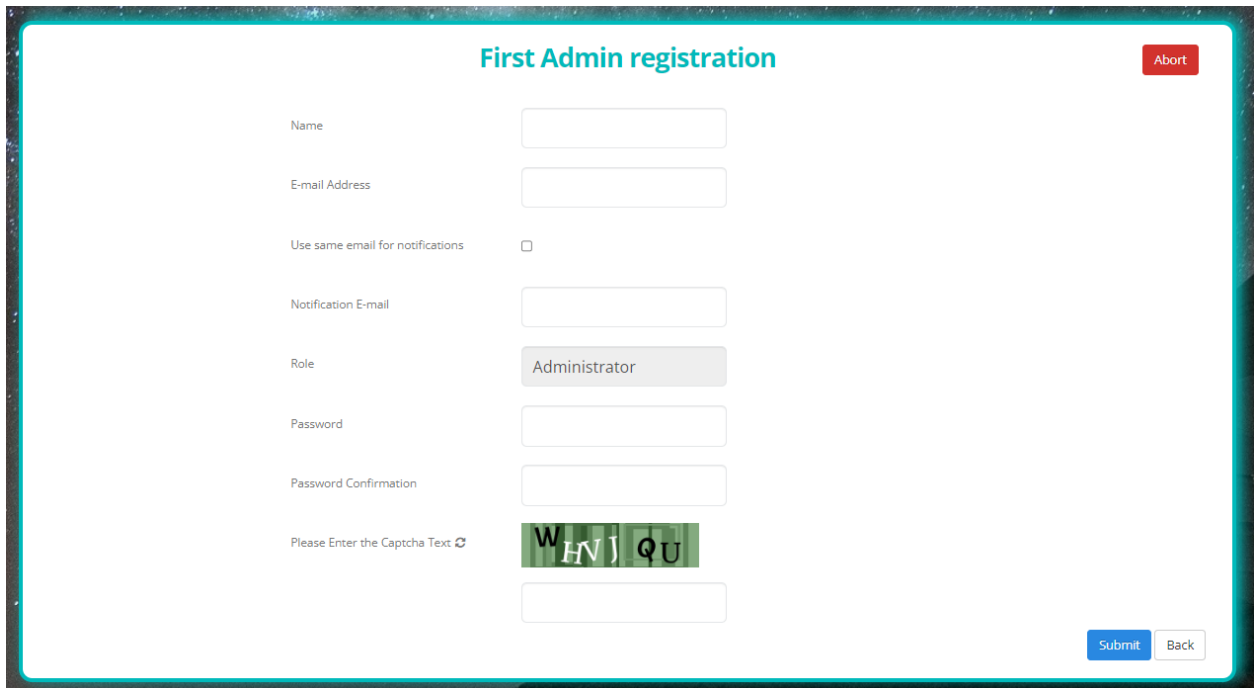
Password (optional)

Submit Back

Figure 5: Mail server configuration stage of AIDA installation.

2.2.3 First Administrator registration

Administrator data must be inserted into the AIDA DB by filling the related form (Figure 6). The user can use its own email address in order to receive AIDA notifications on new user registration and generated reports, or use a different one by acting on the related checkbox.



First Admin registration Abort

Name

E-mail Address

Use same email for notifications ☐

Notification E-mail

Role

Password

Password Confirmation

Please Enter the Captcha Text 

Submit Back

Figure 6: First Admin registration stage of AIDA installation.

Once the form is submitted, if all the inserted data are correct, the user will receive an activation email to confirm its email address (Figure 7). By confirming, AIDA will be opened into a new tab notifying the completion of installation procedure.

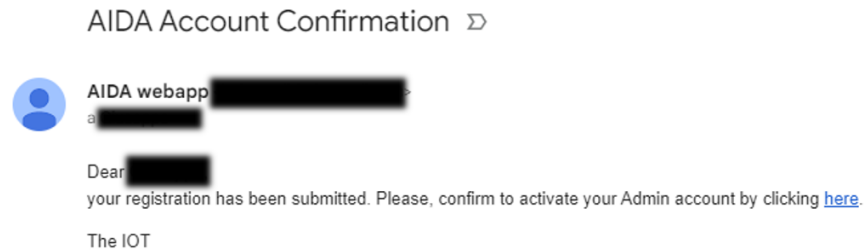


Figure 7: Activation email for the first administrator.

3 User access management

AIDA, as a Rubin-LSST analysis tool, provides access to Vera Rubin/LSST EFD⁶ data through Sasquatch⁷ APIs. This aspect imposes a safe and protected user accounting and access supervision. For this purposes, AIDA foresees two levels of users, accessing with login and password:

- **Administrator(s):** super user(s) with full privileges on the AIDA application, able to exploit all AIDA functionalities and services, but also enabled to monitor and control the activities of all other users, including the possibility to see and manage all user files and running processes (even killing such processes), as well as the responsibility to enable/disable users from the application;
- **Users:** authenticated users, able to exploit all AIDA functionalities and services, but restricted to see and handle only their own files, products and running processes. They performed the registration and obtained the authentication by the administrators.

3.1 Browser compatibility

In terms of web browser compatibility, AIDA has been tested and validated on:

- Chrome v153;
- Microsoft Edge v114;
- Firefox v114;
- Opera v100;

3.2 Registration

In Figure 8 the registration form panel is shown. The procedure designed to register and authenticate AIDA users is the following:

⁶<https://obs-controls.lsst.io/Control-User-Interfaces/EFD.html>

⁷<https://sasquatch.lsst.io/>

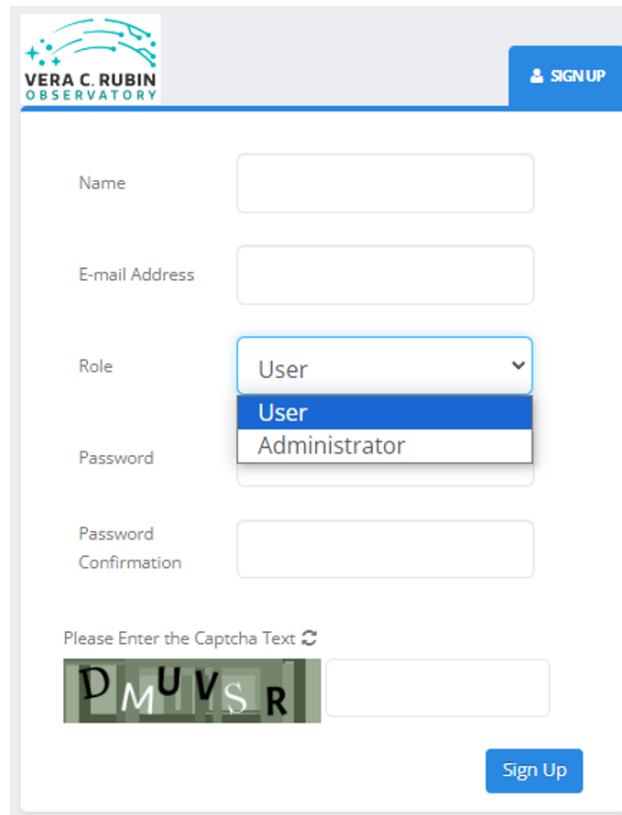


Figure 8: The registration form.

1. From the intro page the interested user opens (through the “Signup” link, Figure 9) and fills the resulting registration form, indicating also the desired level of privileges (user/administrator). Once pressed the “Sign Up” button and only if all form fields have been correctly compiled, the request is delivered and a new record is added in the AIDA local DB with “pending” status, indicating that a new user requested the access authentication, but not yet enabled. The requesting user receives a feedback via e-mail (Figure 10);
2. An e-mail with the registration request is automatically delivered to AIDA administrator(s);
3. AIDA GUI makes available a view panel, available exclusively for administrator level user(s), in which the list of enabled/disabled/pending users is shown. Through this panel, the administrator(s) could manage the authentication requests (see Section 4.2.5 for further details);
4. If and only if the user is enabled by an administrator, the access will be granted, through an automatic e-mail delivered to the user (Figure 11), using the login (e-mail) and pass-

word indicated in the form at the registration time.

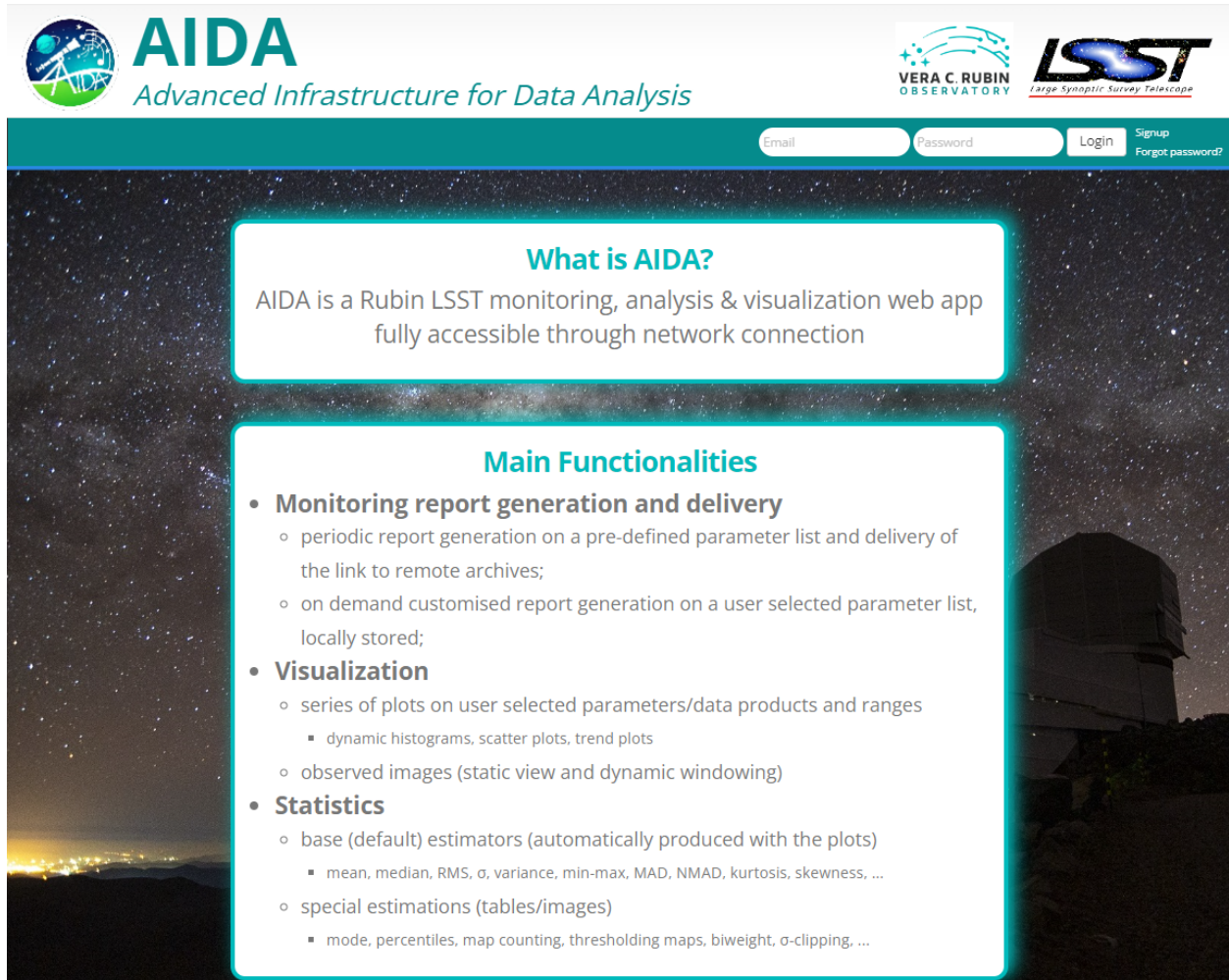


Figure 9: AIDA landing page.

New Registration Request on RUBIN LSST



AIDA admin

a

Dear User,
your registration request has been submitted. You will receive the confirm of your activation as soon as possible.

The IOT

Figure 10: Example of registration feedback sent by AIDA via e-mail.

Your AIDA account has been activated



AIDA admin

Dear Riccio,
your registration has been completed and your account is now active.

Now you can access to your AIDA dashboard by using the password you set during registration.

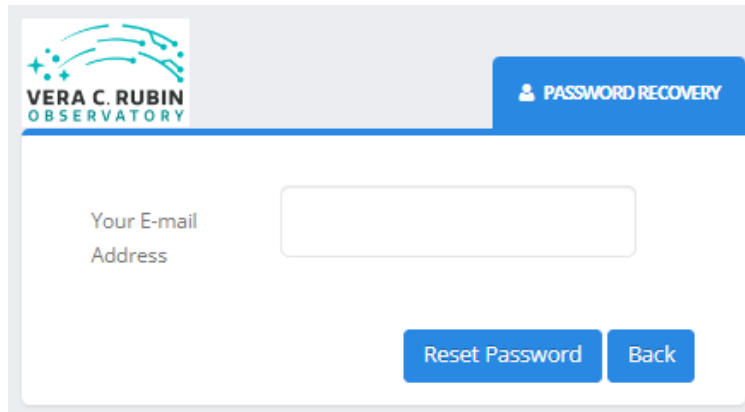
For any issues or feedback, you can send an email to:

The IOT

Figure 11: Example of activation feedback sent by AIDA via e-mail.

3.3 Password Reset

It is possible to reset the password by clicking on the “Forgot password?” link on the landing page (Figure 9), a simple form requiring your email will popup (Figure 12). After that, the user will receive through mail the link to actually reset your password (Figure 13).



The image shows a web interface for password recovery. At the top left is the VERA C. RUBIN OBSERVATORY logo. At the top right is a blue button labeled "PASSWORD RECOVERY" with a user icon. Below the logo, the text "Your E-mail Address" is followed by a text input field. At the bottom right, there are two blue buttons: "Reset Password" and "Back".

Figure 12: Reset password panel.

AIDA Password Recovery



AIDA admin

a me

Dear

Please click on the following link to reset your password.

<http://xxx.xxx.xxx.xxx/aida/recovery.php?key=f2f4012f777983b0fcf4c116ef52f340d466cbbc7e&email=xxxxxxx.xxxxxx@xxxxxxx.xxx&action=reset>

Please be sure to copy the entire link into your browser. The link will expire after 1 day for security reason.

If you did not request this forgotten password email, no action is needed, your password will not be reset. However, you may want to log into your account and change your security password as someone may have guessed it.

Thanks,

The IOT

Figure 13: Example of reset password e-mail sent by AIDA.

4 Main Page

Once logged in, the user dashboard is shown. It consists on two sections:

1. a sidebar to navigate the web app and access to all AIDA functionalities;
2. a number of tabs (4 for users, 6 for administrators), reporting summaries about activities, generated plots, reports, machine learning experiments, and settings.

4.1 Sidebar

AIDA sidebar allows users to navigate the web app and access all AIDA functionalities. It is divided into the following sections (Figure 14):

- **General Info:** in this section the user can find all the information about available systems, parameters and analysis tools (see Section 9);
- **Local Data:** in this section, the user can generate plots from data uploaded from local machine (therefore not interfaced with the remote archives), and to visualize data and experiments stored on the AIDA DB (see Section 10);
- **HKTM:** from this section, the user can generate plots, statistics, and perform machine learning experiments on HKTM data available on remote archives;
- **Images:** images can be uploaded and visualized by the tool in this section or analyzed through some image analysis tools;
- **Reports:** in this section, the user can manage reports functionalities (configuration, running, list of generated reports, and configuration files).

Finally, the installed version of AIDA is reported at the bottom of the sidebar.

4.2 Dashboard

The dashboard shows the operating mode currently active on AIDA, and a number of tabs (4 for users, 6 for administrators), reporting summaries about activities, generated plots, reports, machine learning experiments, and settings:

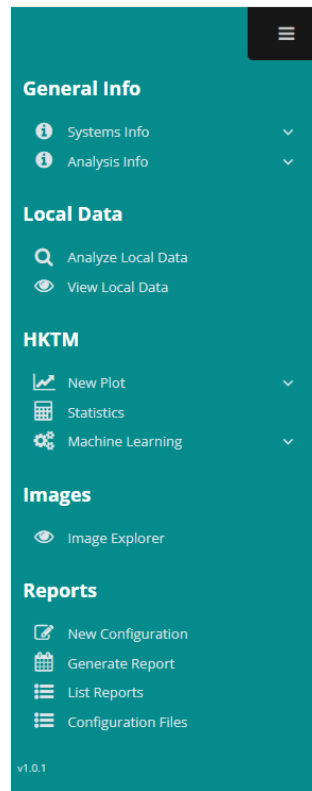


Figure 14: AIDA sidebar.

- Reports
- History
- Stored Experiments
- Machine Learning Experiments
- Administration - Users
- Administration - Settings

4.2.1 Reports tab

Reports tab consists on 4 panels (Figure 15, 16 and 18):

- **Running Reports Generation:** in this panel (Figure 15), the user can see the list of personal reports currently running (administrators can see all reports running). Reports can

be stopped at any time by clicking on the related icon. In addition, administrators can pause and resume periodic reports at any time.

- **Flagged Reports:** this panel (Figure 16) shows the list of flagged reports. By clicking on the info icon, a panel with additional info about the selected report is displayed (Figure 17). For further details about the AIDA flagging system, see Section 11.
- **Available Reports:** In this panel, the tree of all available reports on AIDA are listed, divided for period (Figure 18, left panel);
- **Configuration Files:** In this panel, the tree of all available reports configuration files on AIDA are listed, divided for period (Figure 18, right panel).

Running Reports Generation

10 records per page

Search:

ID	User	Period	Config File	Current Report Start Date	Progress	
13841		ondemand	config_ondemand_long.json	2023-01-01 04:00:00	56.0	
13840		daily	daily_long.json	2023-01-08 04:00:00	73.7	
13839		weekly	weekly.json	2023-01-01 12:00:00	paused	

Showing 1 to 3 of 3 entries

Previous

1

Next

Figure 15: AIDA Dashboard - Reports Tab - Running Reports Generation panel.

Flagged Reports

10 records per page

Search:

Flag	Report ID	Period	Start Date (UTC)	End Date (UTC)	
	IREF_20240419T132520_13840-daily-20230115040000_20230116040000_EFD	daily	2023-01-15 04:00:00	2023-01-16 04:00:00	
	IREF_20240419T132424_13840-daily-20230109040000_20230110040000_EFD	daily	2023-01-09 04:00:00	2023-01-10 04:00:00	
	IREF_20240419T132305_13839-weekly-20230101120000_20230108120000_EFD	weekly	2023-01-01 12:00:00	2023-01-08 12:00:00	
	IREF_20240419T132415_13841-ondemand-20230101040000_20230121040000_EFD	ondemand	2023-01-01 04:00:00	2023-01-21 04:00:00	

Showing 1 to 4 of 4 entries

Previous

1

Next

Figure 16: AIDA Dashboard - Reports Tab - Flagged Reports panel.

4.2.2 History Tab

In this tab (Figure 19), the history of activities on the web app are displayed. The number of records stored and displayed is defined in the web app settings (see Section 4.2.6 for further details). An user can see his/her activities in the “User” panel, while the “Global” panel, showing the activities for all users, is available for administrator(s) only. Further details about a specific

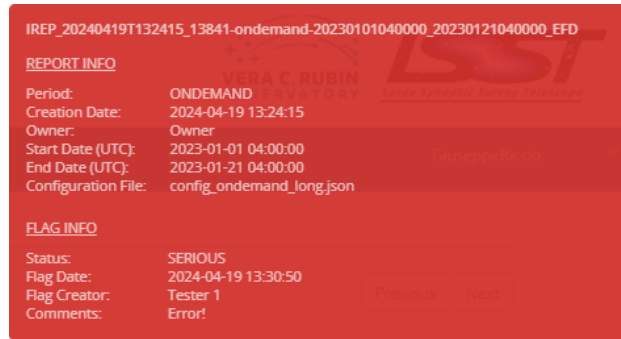


Figure 17: Details on flagged report.

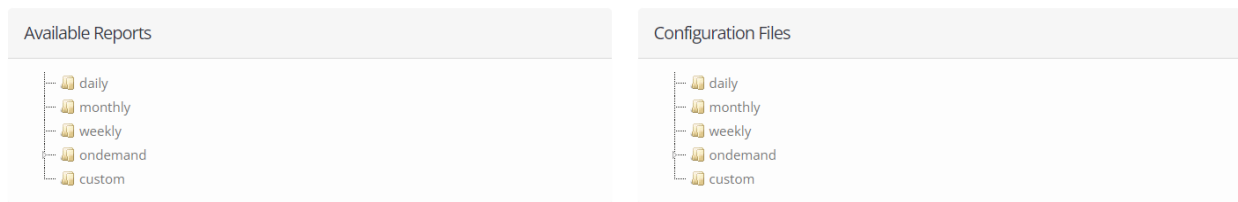


Figure 18: AIDA Dashboard - Reports Tab - Available Reports and Configuration Files panels.

event can be seen by clicking on the related line. In addition, an user/admin can download the complete history (personal and/or, for admin, global) as text file, by using the related button.

4.2.3 Stored Experiments Tab

In the Stored Experiments tab, detailed information about flagged experiments are shown. Experiments are divided in two panels, depending whether the experiment has been flagged as “public” (“Public Archive” panel) or “private” (“Private Archive” panel). Data can be visualized “By Experiment” (Figure 20), i.e. aggregated and ordered by experiment, or “By Parameter” (Figure 21). In the latter case, the list of parameters which have been specifically flagged during an experiment are listed.

In the “By Experiment” visualization, the following info are reported:

- Experiment flag;
- Experiment file name (by clicking it, the experiment pdf is displayed in a separate window)
- Experiment type (plot, report, statistics, image);

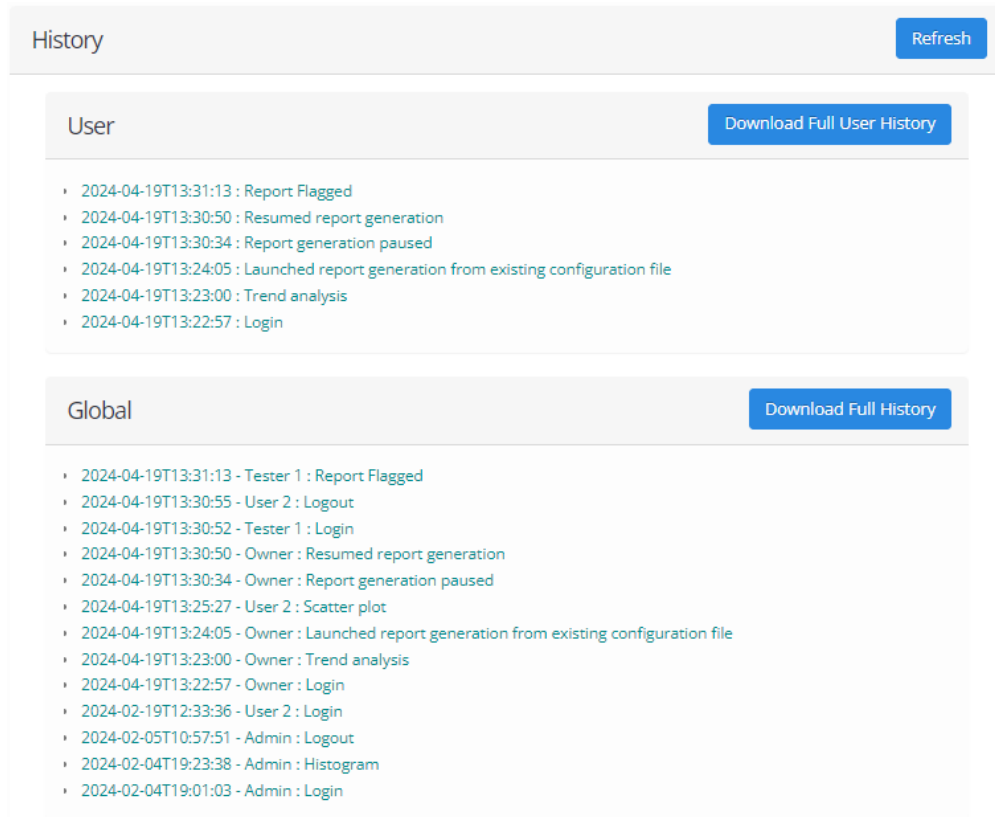


Figure 19: AIDA Dashboard - History Tab.

- Experiment start date and end date (for plots);
- Flag creation date;
- Flag creation user (for “Public” archive);
- Comments, eventually divided for parameter
- a number of available actions:
 - View plot, the stored plot is displayed on AIDA in a separate window;
 - View image, the stored image is displayed on AIDA Image Explorer in a separate window (only available for images from remote archives).

In the “By Parameter” visualization, the following info are reported:

- Parameter flag;

Stored Experiments ☒ By Experiment ☐ By Parameter

Public Archive

10 records per page [Filter columns](#) [Export](#) Search:

Flag	Experiment	Exp Type	Experiment Start Date (UTC)	Experiment Stop Date (UTC)	Generation Date (UTC)	User	Comments	
	IREP_20240419T132415_13841-ondemand-20230101040000_20230121040000_EFD	report	-	-	2024-04-19 13:30:50	GiuseppeRiccio	Error!	
	exp_image_20240419_140155.pdf	image	-	-	2024-04-19 14:02:01	GiuseppeRiccio	External Image example	
	IREP_20240419T132305_13839-weekly-20230101120000_20230108120000_EFD	report	-	-	2024-04-19 13:31:12	GiuseppeRiccio	Warning comment	
	exp_trend_20240419_140026.pdf	plot	2023-01-01 00:00:00	2023-01-15 00:00:00	2024-04-19 14:00:27	GiuseppeRiccio	SysA.temperature: Test plot OK	
	IREP_20240419T132424_13840-daily-20230109040000_20230110040000_EFD	report	-	-	2024-04-19 13:30:34	GiuseppeRiccio	All OK	
	IREP_20240419T132520_13840-daily-20230115040000_20230116040000_EFD	report	-	-	2024-04-19 13:30:06	GiuseppeRiccio	-	
	exp_statistics_20231205_100403.pdf	statistics	2023-01-02 00:00:00	2023-01-03 00:00:00	2023-12-05 10:04:03	GiuseppeRiccio	-	

Showing 1 to 7 of 7 entries

Previous **1** Next

Filters [Reset](#)

by Flag by Experiment by Exp Type

By Start Date By End Date By Experiment Date By User

Figure 20: AIDA Dashboard - Stored Experiments Tab - Public Archive sorted by Experiment.

Personal Archive

10 records per page [Filter columns](#) [Export](#) Search:

Flag	Parameter	System	Experiment	Exp Flag	Experiment Start Date (UTC)	Experiment Stop Date (UTC)	Generation Date (UTC)	Comments
	SysA.temperature	EFD	exp_trend_20240221_124606.pdf		2023-01-01 00:00:00	2023-01-14 00:00:00	2024-02-21 12:46:06	
	SysB.current	EFD	exp_statistics_20231201_141239.pdf		2023-01-02 00:00:00	2023-01-03 00:00:00	2023-12-01 14:12:39	

Showing 1 to 2 of 2 entries

Previous **1** Next

Filters [Reset](#)

by Flag By Parameter By System

by Experiment by Exp Flag

By Start Date By End Date By Experiment Date

Figure 21: AIDA Dashboard - Stored Experiments Tab - Personal Archive sorted by Parameter.

- Parameter name;
- Parameter system
- Experiment file name (by clicking it, the experiment pdf is displayed in a separate window);
- Experiment flag;
- Experiment start date and end date (for plots);

- Flag creation date;
- Flag creation user (for “Public” archive);
- Comments.

4.2.4 Machine Learning Experiments Tab

In the Machine Learning Experiments tab, detailed information about performed ML experiments is shown. ML experiments executed by all the users are displayed in a panel that reports technique and model used, data source, data interval, user who ran the experiment and date of execution. For each experiment, it is possible to download the model file (in *joblib* format), the CSV output and the log (Figure 22).

Machine Learning Experiments							
10 records per page		Search:					
ID	ML Technique	Model	Data Source	Dates Interval	User	Generation Date	
1	Clustering	KMeans	FAKE	[2023-01-11 00:00:00, 2023-01-12 00:00:00]	GiuseppeRiccio	2026-07-20 10:51:04	  
2	Classification	DecisionTreeClassifier	FAKE	[2023-01-11 00:00:00, 2023-01-12 00:00:00]	GiuseppeRiccio	2026-07-20 10:55:11	  
3	Regression	DecisionTreeRegressor	FAKE	[2023-01-11 00:00:00, 2023-01-12 00:00:00]	GiuseppeRiccio	2026-07-20 10:55:53	  

Figure 22: AIDA Dashboard - Machine Learning Experiments Tab.

4.2.5 Administration - Users tab

Through this exclusive tab (Figure 23), the administrator(s) could monitor and manage users accesses and registration requests, through 3 panels:

- “Pending Requests” panel: accessing to this panel, administrator(s) can see all pending registration requests, and confirm or reject them;
- “Active Users” panel shows info about active users, giving the possibility to administrator(s) to temporarily deactivate or definitely remove an user;
- “Deactivated Users” panel shows the list of temporarily deactivated users, giving the possibility to administrator(s) to re-activate or definitely remove them.

Pending Requests

10

▼ records per page

Search:

Username	E-Mail	Role	Request Date	
[REDACTED]	[REDACTED]	user	2022-12-19 13:43:15	✓ 🗑

Showing 1 to 1 of 1 entries

Previous1Next

Active Users

10

▼ records per page

Search:

Username	E-Mail	Role	Activation Date	Last Login	Last Logout	
[REDACTED]	[REDACTED]	admin	2022-12-19 13:43:15	2024-04-19 13:11:56	2024-04-12 11:51:56	STOP 🗑
[REDACTED]	[REDACTED]	user	2024-04-12 11:44:43	-	-	

Showing 1 to 2 of 2 entries

Previous1Next

Deactivated Users

10

▼ records per page

Search:

Username	E-Mail	Role	Deactivation Date	
[REDACTED]	[REDACTED]	user	2024-04-12 11:05:44	✓ 🗑

Showing 1 to 1 of 1 entries

Previous1Next

Figure 23: The supervision panel of administrator(s).

4.2.6 Administration - Settings tab

In this administrators' exclusive tab (Figure 24), three panels are available, each one devoted to specific aspects of AIDA configuration:

- "Operating Modes": administrator(s) can change operating mode, choosing among "Nominal", "Contingency" and "Commissioning". By changing the current operating mode, AIDA will automatically connect to the related repository⁸;
- "Systems": in this panel all the systems available for the current operating mode are listed. Administrator(s) have the possibility, for each system, to activate/deactivate a specific kind of monitored data;
- "Web App Settings": through this panel, administrator(s) can change AIDA main settings, configure a new SMTP server to use or export existing data for a future new brand installation (see Section 2.2). By clicking on the related button, a modal window will open in which the user can proceed with the selected operation.

⁸For Vera Rubin/LSST, all the operating modes points to the same configuration.

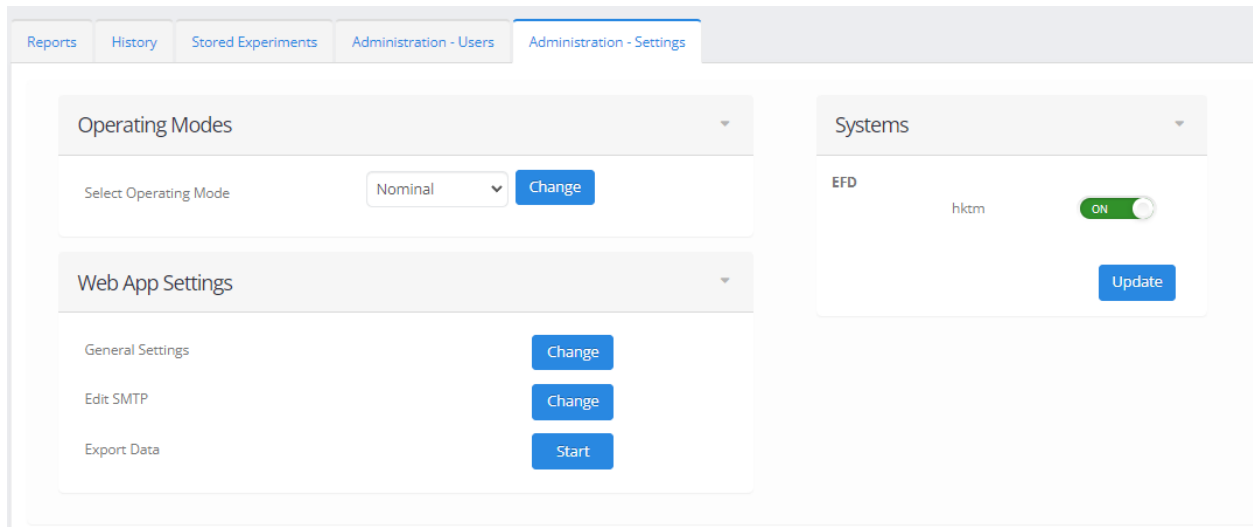
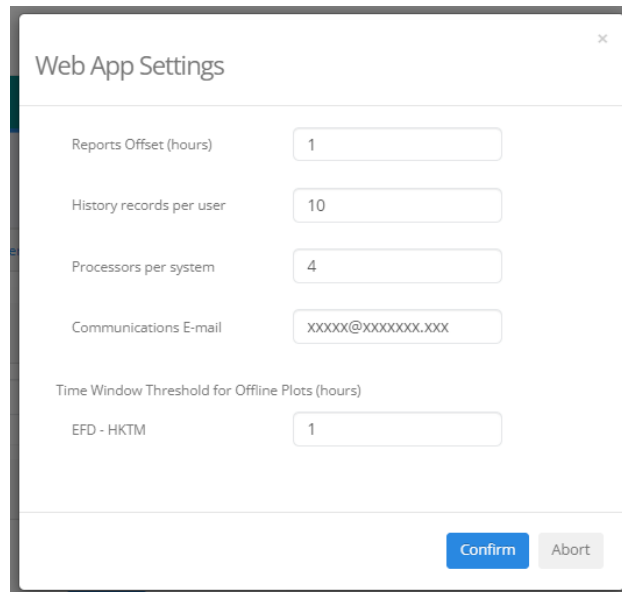


Figure 24: Administration - Settings tab.

General Settings

Available settings in “General Settings” panel are reported in Figure 25:

- **Report Offset:** time to wait, since start date, before starting data acquisition for report generation (it depends on the time delay to have data available on the remote repository since telescope’s acquisition);
- **History records per user:** number of history records to store into AIDA DB for each user;
- **Processors per system:** number of processor to use for parallel analysis for each system (it depends on the machine hosting AIDA)
- **Communications email:** One or more email addresses to use to notify new user requests or report generation (see Section 2.2 for further details);
- **Time Window Threshold for Offline Plots:** in order to avoid the user waiting plot generation for a long time because of the huge amount of data to analyze, AIDA can perform the analysis offline, then notifying its completion by e-mail to the user. This setting defines the maximum time window for a plot to be generated online; values higher than this will drive AIDA to generate plots offline. If set to 0, then the experiment will be performed exclusively online for the related usecase.



The image shows a 'Web App Settings' modal window with a close button (X) in the top right corner. It contains several configuration fields:

- Reports Offset (hours): 1
- History records per user: 10
- Processors per system: 4
- Communications E-mail: XXXXX@XXXXXXXXX.XXX
- Time Window Threshold for Offline Plots (hours): 1
- EFD - HKTM: 1

At the bottom right, there are two buttons: 'Confirm' (blue) and 'Abort' (grey).

Figure 25: General settings modal window.

Edit SMTP

The “Edit SMTP” modal panel includes the same form shown in Figure 5 and the procedure to update the SMTP server settings is the same as the one described in Section 2.2.

Export Data

By starting the “Export Data” procedure, the modal panel of Figure 26 will be shown. In such a panel, the administrator(s) can select which info to export. By confirming the export, a tar.gz file will be created and downloaded, containing all AIDA configuration and setting files, stored experiments files and AIDA DB tables required to restore data. This file could be used to import stored data into a new AIDA configuration (see Section 2.2). The tar.gz archive includes text files (pdf, configuration, user folders etc.), related to the user’s operation performed during working session. No retrieved data will be stored in this archive, since all data are automatically deleted at the end of the user’s session. Therefore, the dimension of such an archive will depend on the user’s operations, but it will never reach critical sizes (maximum few GB in the worst case).

NOTE: Any modification of this file could cause SERIOUS malfunctioning and/or data loss.

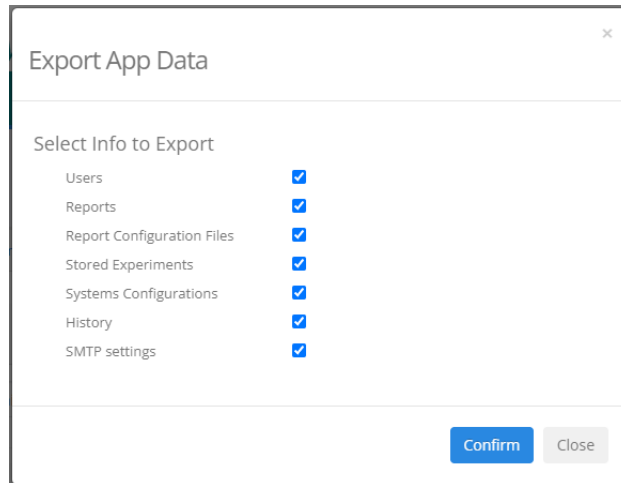


Figure 26: Export data modal window.

5 Plots generation

Current types of standard plots available, under left menu "HKTM" section, are 2D histograms (option "Histogram"), scatter plots (option "Scatter Plot") and trend analysis plots (option "Trend Analysis"). Each of them foresees a setup form, specifically designed for each available system, type of plot and operating mode.

5.1 Standard plots

All the standard plots (trend, scatter and histogram) require to be configured through online forms. Even though forms are specific for each kind of plot, some fields are in common among them:

- **Data source:** the involved system;
- **Parameter:** the parameter to monitor;
- **Date range (UTC) :** start date and end date of plotted data.

Main parameters to plot are identified as "Y axis parameters" (for trend and scatter plots) or "Parameters" (for histogram).

In addition, for a scatter plot, the "X" parameter has to be configured (fields are the same as for "Y" parameter), while the user has to define the bin size (or the number of bins) for a histogram.

To completely characterize parameters, one or more additional fields are present (for example, the specific subsystem to which the parameter refers).

Finally, the user can optionally select which statistical analysis to produce together the plot, by clicking on "Advanced Statistics", otherwise a basic statistics will be provided (see Section 6 for further details).

Some examples of such setup forms for each plot type are shown, respectively, in Figure 27 for trend analysis, Figure 28 for scatter plots and Figure 29 for histograms.

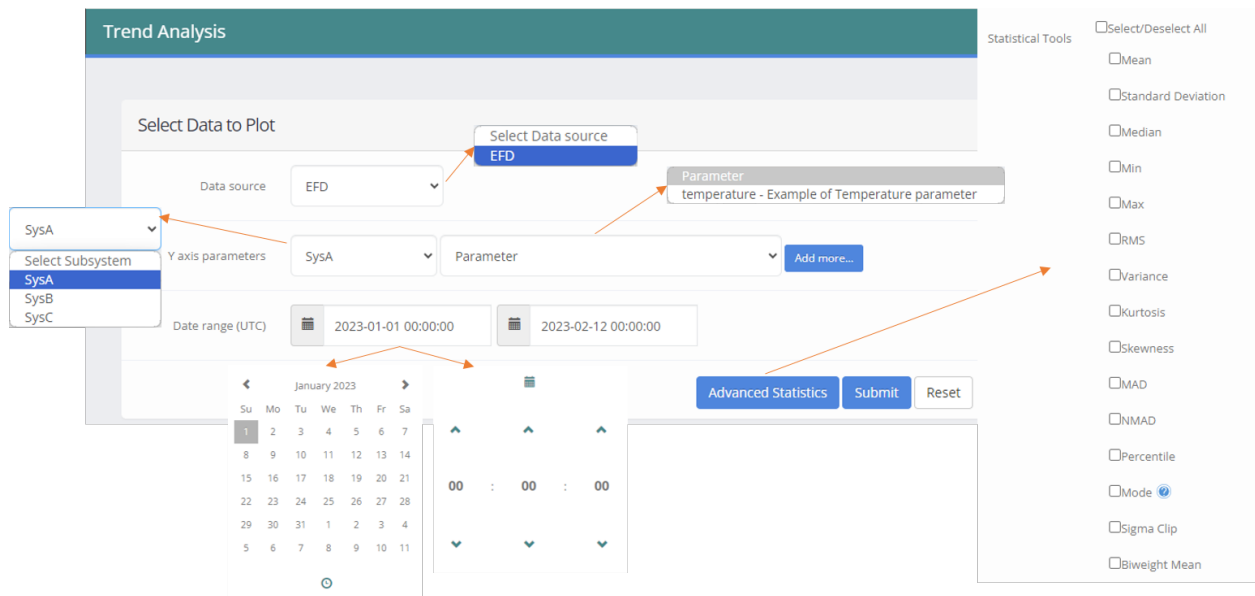


Figure 27: The form panel for the setup of a trend analysis task.

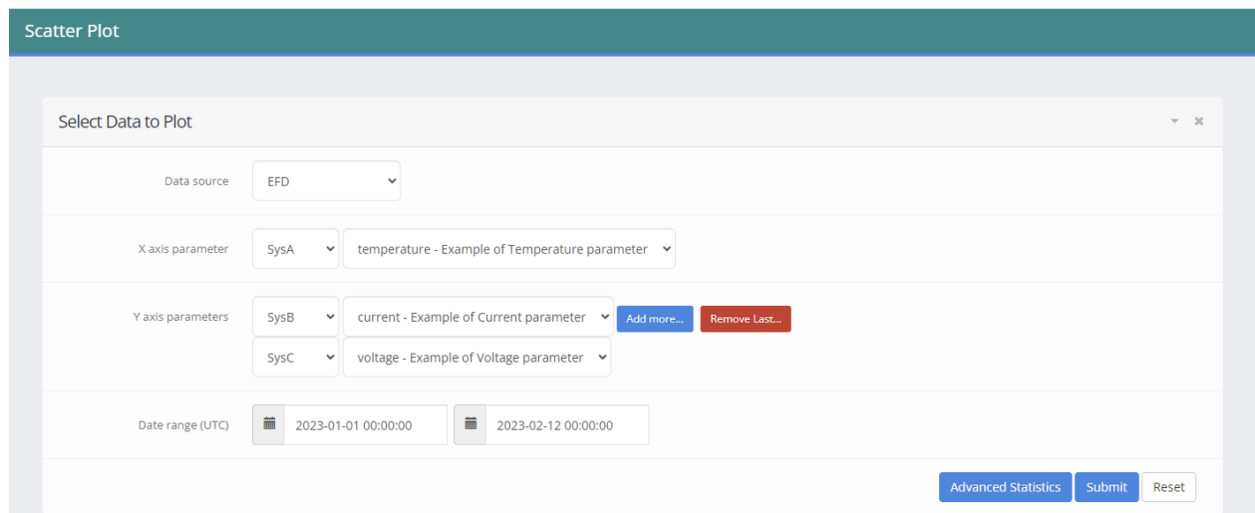


Figure 28: The form panel for the setup of a scatter plot.

The results of the form submission are shown in two tabs:

1. plot displaying data points;
2. statistical analysis results (defaults or advanced) for each involved parameter;

If the range between start and end date is larger than the "Time Window Threshold for Offline

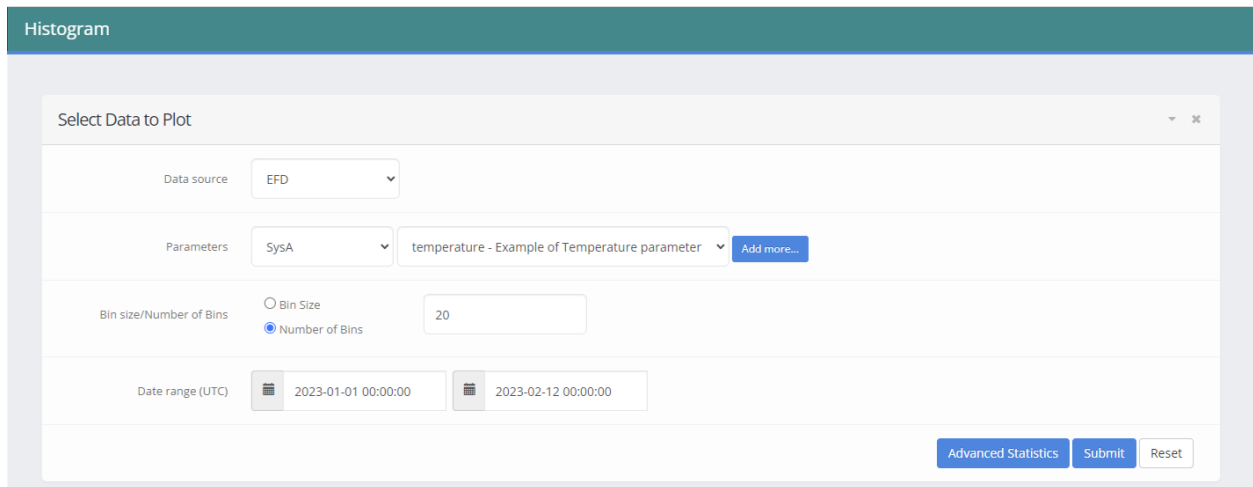


Figure 29: The form panel for the setup of a histogram.

Plots (hours)" settings for the specific case (Figure 25), AIDA will show a message, communicating that the plot will be generated offline. Once the plot is generated, the user will receive an e-mail reporting the summary of the experiment and the link to visualize the plot (Figure 30)

New Trend Analysis generated



A new data analysis has been performed by AIDA:

SUMMARY

Experiment : Trend Analysis

Data Origin : HKTM - EFD

Y Parameter(s) : SysB.current,SysC.voltage

Date Range : 2023-01-01 00:00:00 - 2023-01-14 00:00:00

Statistics : Mean, Standard Deviation, Median, Min, Max, RMS, Variance, Kurtosis, Skewness, MAD, NMAD

Click [here](#) to view results.

Figure 30: Example of e-mail sent by AIDA after offline plot generation.

An example of generated plots is shown, respectively, in Figure 31 for trend analysis, Figure 32 for scatter plot and Figure 33 for the histogram.

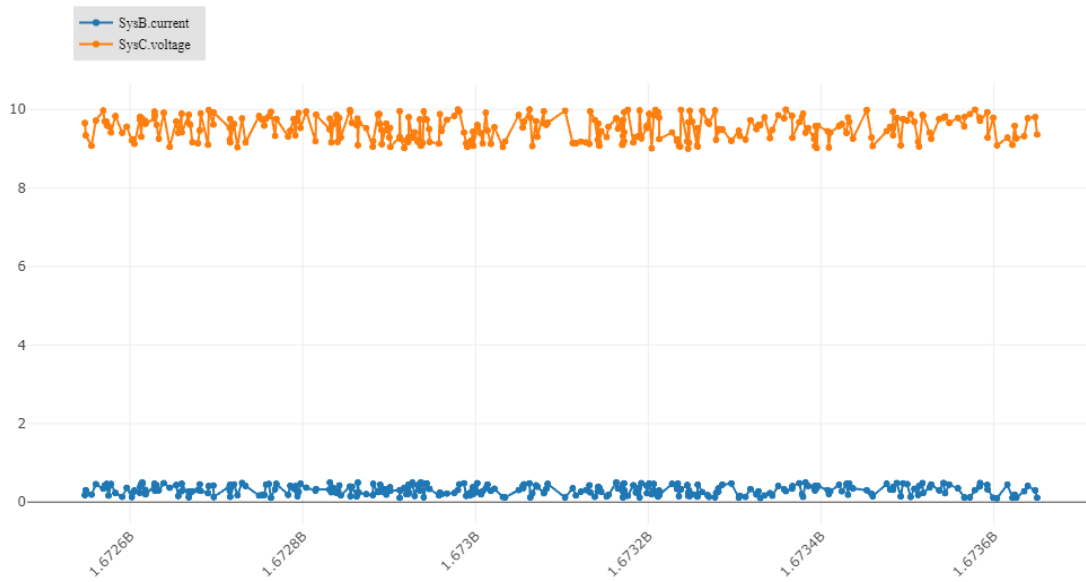


Figure 31: Example of trend analysis plot.

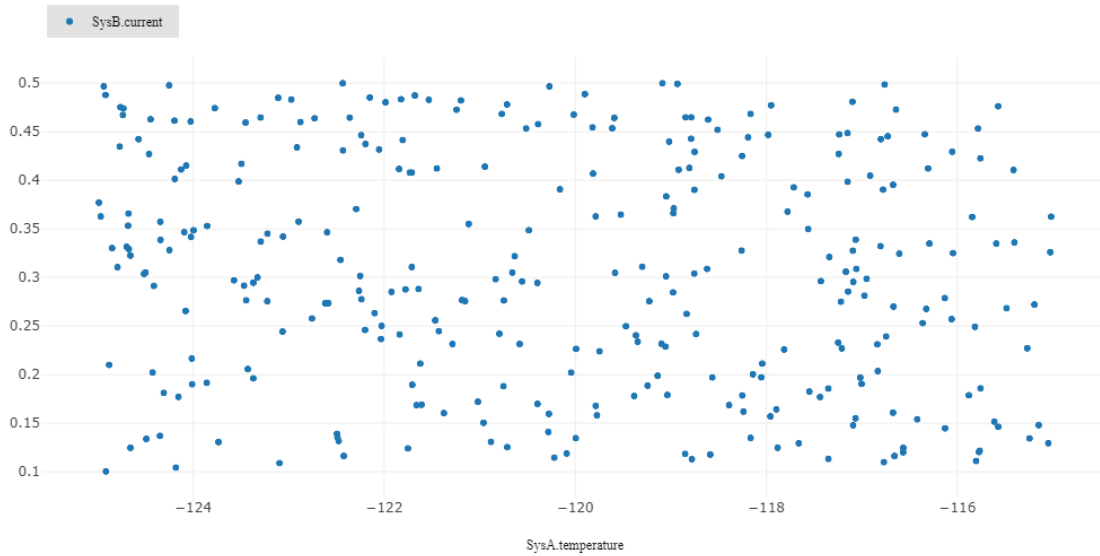


Figure 32: Example of scatter plot.

5.2 Plot navigation bar

Once a plot is generated, it is possible to execute some operations to navigate into it, to generate some products and to customize plot graphics, through a navigation bar in the top-right

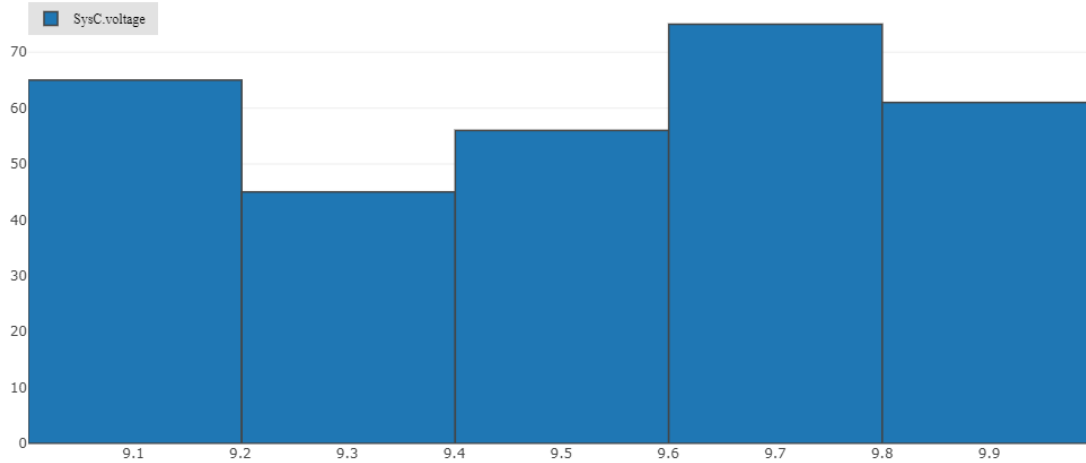


Figure 33: Example of histogram.

corner of the plot (Figure 34). With reference to Figure 34, from left to right, the user can find the following buttons:

- **Download Plot as Image:** the plot is downloaded as PNG image, as it is visualized;
- **Save experiment as PDF:** the plot and the statistical analysis are downloaded as a PDF report file, after optionally having assigned a flag to the plot (see Section 11 for further details on the flagging system);
- **Save results as CSV:** points displayed into the plot are downloaded as a CSV file;
- **Zoom:** the plot can be zoomed, by selecting a specific area or rolling the mouse wheel;
- **Pan;**
- **Box Select:** the user can highlight a specific area of the plot;
- **Zoom in;**
- **Zoom out;**
- **Autoscale:** the plot is auto-scaled to display all the points;

- **Reset axes:** the plot axes are restored to default settings;
- **Toggle Spike Lines:** spike lines are toggled when the mouse is over a point;
- **Show closest data on hover:** when the mouse is over a point, the related pair of values (X,Y) is displayed;
- **Compare data on hover:** when the mouse is over a point, all the values Y corresponding to the related X are displayed;
- **Customize Plot:** a side panel is displayed to customize plot graphics;



Figure 34: Navigation bar for plots graphics customization.

6 Statistical analysis

Currently, the available statistical estimators are:

- Mean, Median;
- Min, Max;
- Mode: if the precision is greater than 0, data will be arranged into bins of width = $2 \times \text{precision}$ and the median of values in each bin is taken for calculation. Otherwise the single values will be considered;
- RMS, Standard Deviation, Variance;
- Kurtosis, Skewness, Percentiles;
- MAD, NMAD: Median Absolute Deviation and the normalized version ($1.4826 \times \text{MAD}$);
- Sigma Clip, defined as a way to avoid the effect of outliers, which could significantly bias the measurement of noise statistics. This estimator performs an iterative process removing all data distribution points lower or higher than $f + \alpha\sigma$, where f is the function used to compute the center value and σ the function used to clip the data distribution, while α is the multiplicative constant parameter. Currently, during configuration it is possible to set f and α ;
- Robust mean, based on biweight⁹, a transformation used in robust analysis. It combines the properties of statistical robustness with relatively high efficiency. Here, robustness means that changes in a small portions of data do not cause large changes in the mean estimate.

Statistical analysis can be accessed by clicking "Statistics" on the left menu in the dashboard. The user has to configure the experiment by filling the usual form on parameters selection and checking the statistical estimators to compute among the listed ones. The same estimators list is displayed when the "Advanced Statistics" button on plot configuration forms is clicked.

⁹"Data Analysis and Regression" Mosteller and Tukey, Addison-Wesley, 1977

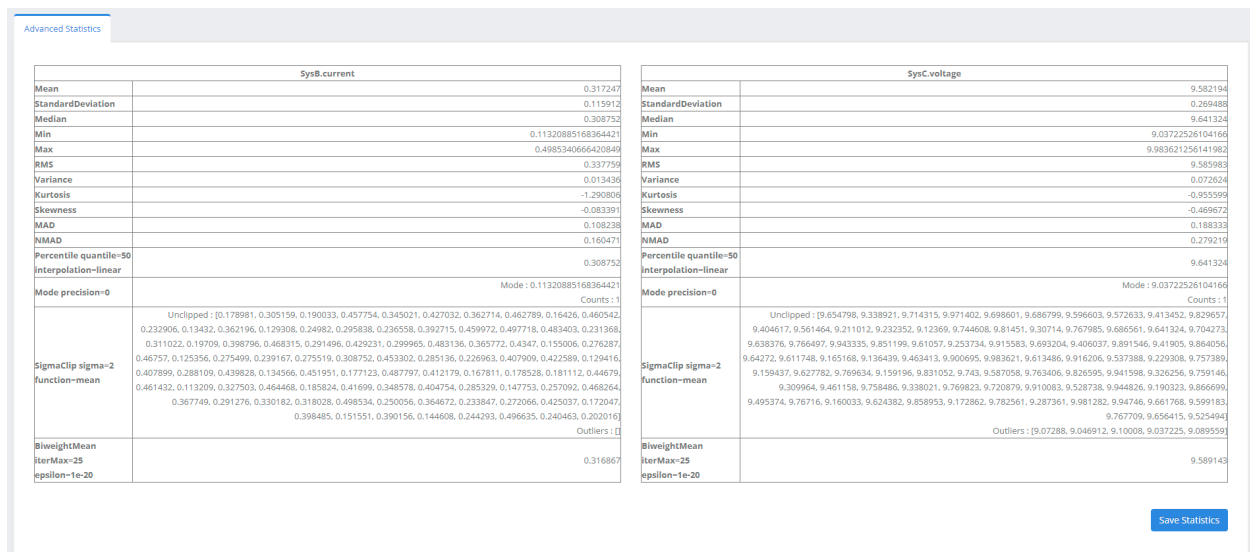


Figure 35: Example of advanced statistical analysis.

7 Machine Learning integrated tool

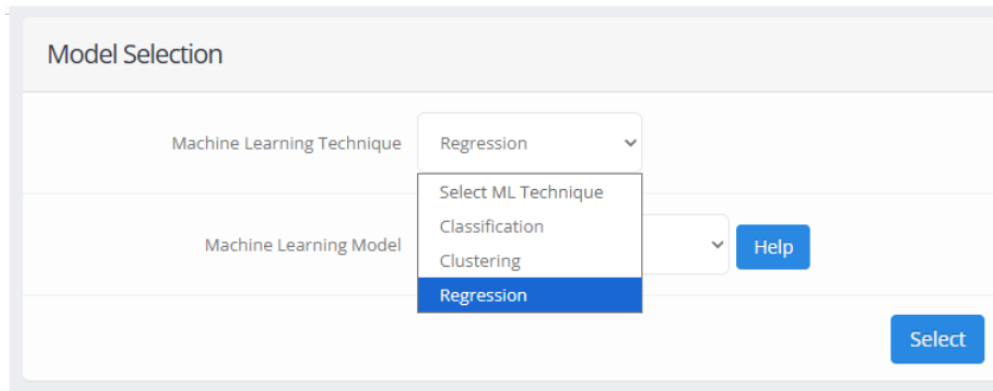
According to the wide and always increasing use of data-driven methods to handle and explore huge amounts of multi-dimensional data, a specific analytics module, based on machine learning paradigms and models, is integrated within AIDA.

All data available in remote repositories can be used to train/test some classification or regression models, or as input for clustering algorithms, by filling out a form accessible by clicking "*Machine Learning -> Train a model*" in the sidebar. The ML models available on AIDA are based on the Python package **scikit-learn**¹⁰.

The experiment configuration is a two-step process:

1. **Model Selection:** the user defines the ML Technique (*Clustering, Classification or Regression*) through a drop-down menu (Figure 36). By changing the value of this cell, a second drop-down menu will appear with the list of available ML models that can be used (Figure 37). An "Help" page is available for each model by clicking the related button (Figure 38).
2. **Data and Model configuration:** in the second step, the user has to define the data source (Figure 39) and the hyper-parameters setup (Figure 40). The data selection is similar to the one for plots (see Section 5) with the difference that the fields labels are "*Labels*" and "*Features*" instead of "*X axis*" and "*Y axis*". Moreover, the choice of the "*Labels*" parameter is limited for classification and regression experiments only. **NOTE:** only integer values are allowed for a parameter chosen as label in a classification experiment.

¹⁰<https://scikit-learn.org/stable/>



Model Selection

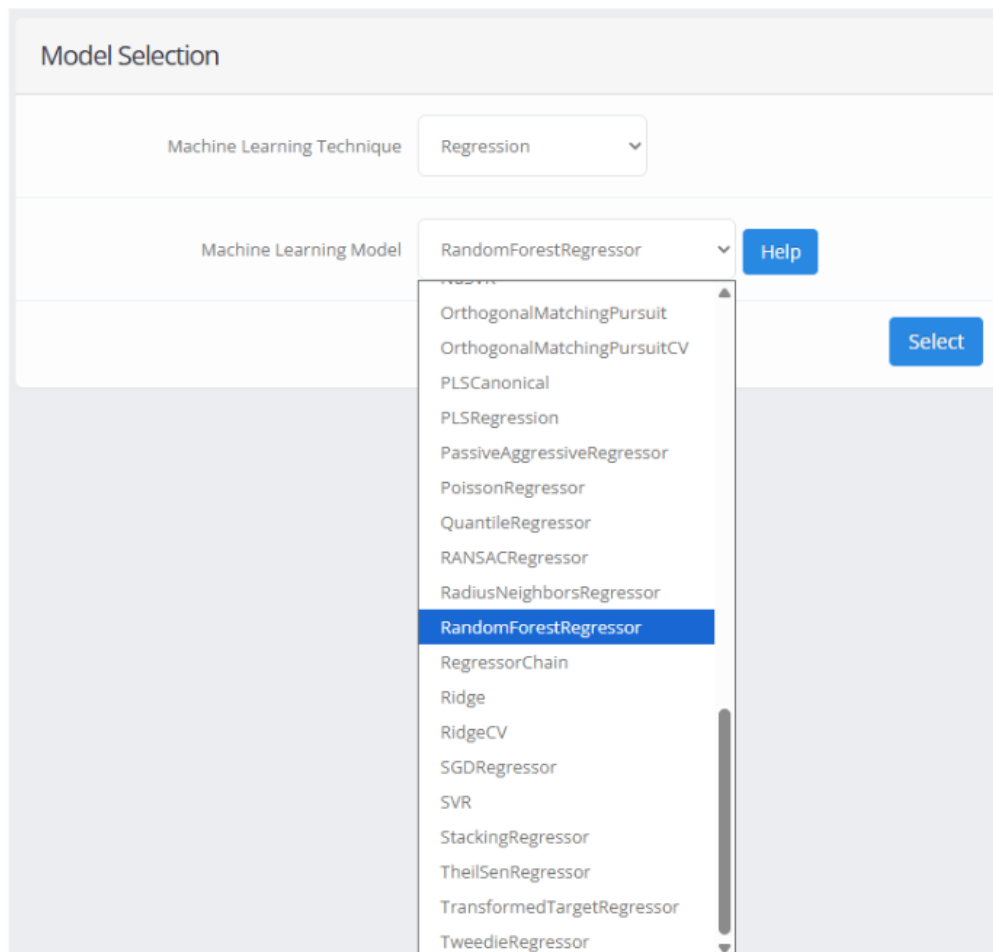
Machine Learning Technique: Regression

Machine Learning Model: [dropdown]

Help

Select

Figure 36: First step of a ML experiment configuration: Technique choice



Model Selection

Machine Learning Technique: Regression

Machine Learning Model: RandomForestRegressor

Help

Select

Figure 37: First step of a ML experiment configuration: Model choice

Help - RandomForestRegressor

Help on class RandomForestRegressor in module sklearn.ensemble._forest:

```

class RandomForestRegressor(ForestRegressor)
|   RandomForestRegressor(n_estimators=100, *, criterion='squared_error', max_depth=None, min_samples_split=2,
|   |
|   | A random forest regressor.
|   |
|   | A random forest is a meta estimator that fits a number of decision tree
|   | regressors on various sub-samples of the dataset and uses averaging to
|   | improve the predictive accuracy and control over-fitting.
|   | Trees in the forest use the best split strategy, i.e. equivalent to passing
|   | 'splitter="best"' to the underlying :class:`~sklearn.tree.DecisionTreeRegressor`.
|   | The sub-sample size is controlled with the 'max_samples' parameter if
|   | 'bootstrap=True' (default), otherwise the whole dataset is used to build
|   | each tree.
|   |
|   | For a comparison between tree-based ensemble models see the example
|   | :ref:`sphx_glr_auto_examples_ensemble_plot_forest_hist_grad_boosting_comparison.py`.
|   |
|   | Read more in the :ref:`User Guide`.
|   |
|   | Parameters
|   | -----
|   | n_estimators : int, default=100
|   |     The number of trees in the forest.
|   |
|   | .. versionchanged:: 0.22
|   |     The default value of ``n_estimators`` changed from 10 to 100
|   |     in 0.22.
|   |
|   | criterion : {"squared_error", "absolute_error", "friedman_mse", "poisson"}, default="squared_error"
|   |     The function to measure the quality of a split. Supported criteria
|   |     are "squared_error" for the mean squared error, which is equal to
|   |     variance reduction as feature selection criterion and minimizes the L2
|   |     loss using the mean of each terminal node, "friedman_mse", which uses
|   |     mean squared error with Friedman's improvement score for potential
|   |     splits, "absolute_error" for the mean absolute error, which minimizes
|   |     the L1 loss using the median of each terminal node, and "poisson" which
|   |     uses reduction in Poisson deviance to find splits.
|   |     Training using "absolute_error" is significantly slower
|   |     than when using "squared_error".
|   |
|   | .. versionadded:: 0.18
|   |     Mean Absolute Error (MAE) criterion.
|   |
|   | .. versionadded:: 1.0
|   |     Poisson criterion.
|   |
|   | max_depth : int, default=None
|   |     The maximum depth of the tree. If None, then nodes are expanded until
|   |     all leaves are pure or until all leaves contain less than
|   |     min_samples_split samples.
|   |
|   | min_samples_split : int or float, default=2
|   |     The minimum number of samples required to split an internal node:

```

Figure 38: First step of a ML experiment configuration: Model Help page.

Machine Learning

Data source: EPD

Label: MTM2, temperature, exhaust0 - Exhaust temperatures: #1 and #2

Features:

Feature	Type	Description
MTM2	temperature	ring0 - Ring temperatures: LG2-1, LG2-2, LG2-3, LG2-4, LG3-1, LG3-2, LG3-3, LG3-4, LG4-1, LG4-2, LG4-3, and LG4-4.
MTM2	temperature	ring1 - Ring temperatures: LG2-1, LG2-2, LG2-3, LG2-4, LG3-1, LG3-2, LG3-3, LG3-4, LG4-1, LG4-2, LG4-3, and LG4-4.
MTM2	temperature	ring2 - Ring temperatures: LG2-1, LG2-2, LG2-3, LG2-4, LG3-1, LG3-2, LG3-3, LG3-4, LG4-1, LG4-2, LG4-3, and LG4-4.
MTM2	temperature	ring3 - Ring temperatures: LG2-1, LG2-2, LG2-3, LG2-4, LG3-1, LG3-2, LG3-3, LG3-4, LG4-1, LG4-2, LG4-3, and LG4-4.
MTM2	temperature	ring0 - Ring temperatures: LG2-1, LG2-2, LG2-3, LG2-4, LG3-1, LG3-2, LG3-3, LG3-4, LG4-1, LG4-2, LG4-3, and LG4-4.

Date range (UTC): 2026-05-01 00:00:00 to 2026-05-01 01:00:00

Figure 39: Second step of a ML experiment configuration: selection of the input tabular data

Configuration of RandomForestRegressor

n_estimators	100
criterion	squared_error
max_depth	None
min_samples_split	2
min_samples_leaf	1
min_weight_fraction_leaf	0.0
max_features	1.0
max_leaf_nodes	None
min_impurity_decrease	0.0
bootstrap	True
oob_score	False
n_jobs	None
random_state	None
verbose	0
warm_start	False
ccp_alpha	0.0
max_samples	None
monotonic_cst	None

Train - Test Split: Percentage of data to be used as Train: 70

Random Seed for the Split: None

Figure 40: Second step of a ML experiment configuration: configuration of the model hyper-parameters

Once the experiment is submitted by clicking the appropriate button, it begins: a clustering experiment will assign a cluster to each point of the dataset, while in the case of a classification or regression experiment, the dataset will be automatically and randomly split into train and test sets, according with the configuration of the model hyper-parameters, and the two phases will be carried out one after the other. As with plots and statistics experiments, if the date interval exceeds the defined "Time Window Threshold" (see Section 4.2.6), the experiment will run in offline mode, and an email will notify of its completion, providing a link to the results page; otherwise the results page will be displayed when finished once processing is complete.

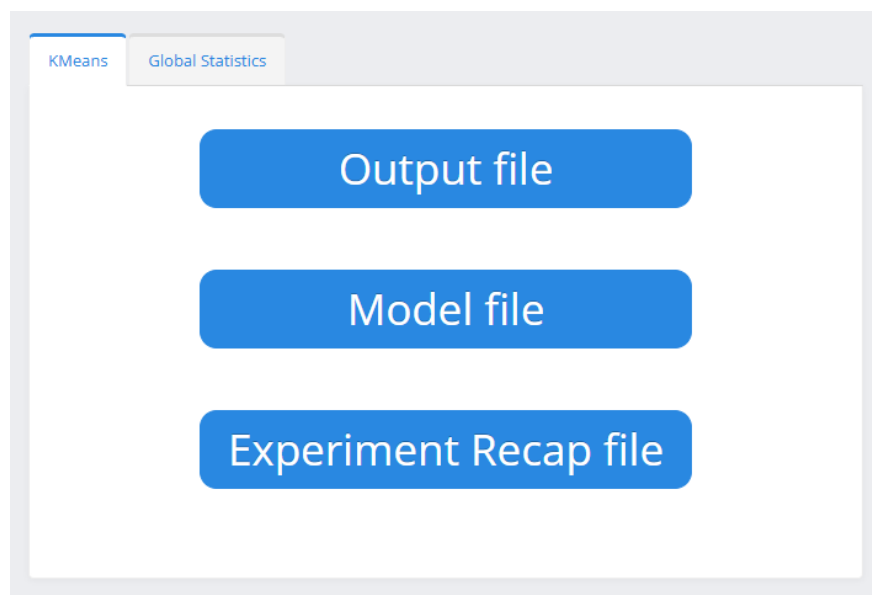


Figure 41: Page collecting the results of a ML experiment.

The results page shows two tabs: the first tab features three buttons, each designed for downloading one of the expected outputs (Figure 41) :

- the **Output file** allows to download a CSV file, reporting different info for each data record: an identifier of the record (usually the acquisition date), the belonging cluster in the case of a clustering experiment, or, for regression and classification, the target and output values of the records in the test set, and their difference;
- the **Model file** is a *joblib* file that can be applied on new data by an external Python code based on the scikit-learn package;
- the **Experiment Recap file** reports info about the configuration of the experiment, as

defined in the configuration form.

The second tab shows the standard statistical indicators for the parameters chosen for the experiment.

All the experiment info and outputs are locally stored on AIDA and are listed in the *Machine Learning Experiments* tab on the Dashboard (see Section 4.2.4).

8 Image Explorer

In terms of statistical evaluation of the loaded images, the Image Explorer tool is able to load and display any image and to dynamically navigate into it. It provides a number of advanced features, such as image statistics, tiling, blinking, geometric markers, arbitrary zoom, rotation, pan, and a variety of coordinate systems. making also possible to select an arbitrary sub-region, including borders, to get coordinate ranges and export any kind and shape of thumbnails. The overall view of the Image Explorer is shown in Figure 42.

The AIDA Image Explorer tool is based on JS9 Javascript library¹¹ (v3.6.2): refer to the official user manual ¹² for a description of general features.

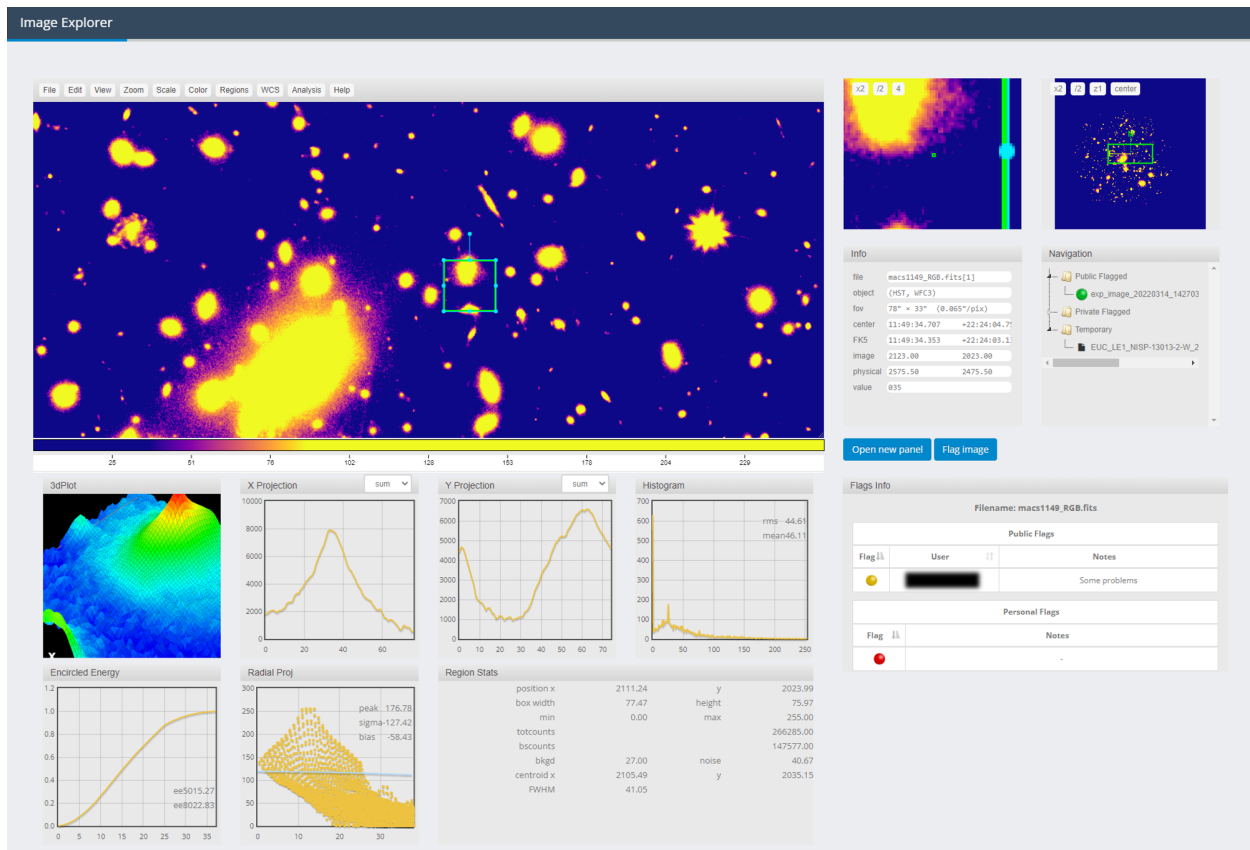


Figure 42: Image Explorer overall view.

The tool can be accessed by the left menu (Figure 14).

¹¹<https://js9.si.edu>

¹²<https://js9.si.edu/js9/help/user.html>

The Image Explorer consists of several panels, displaying image details, info and analysis results (Figure 42):

1. **Top-left panel:** the image container, displaying by default the central region of the image (if it is too large to be fully displayed);
2. **Top-right panel:** four mini-panels (from top-left, clockwise):
 - **Panner**, dynamically displaying the image region centred on the mouse position;
 - **Magnifier**, which displays the whole image, highlighting the region shown in the main panel;
 - **Navigation**, showing images already stored on AIDA (flagged or temporary) which can be loaded on click;
 - **Info**, about the displayed image and the pixel dynamically pointed by the mouse;
3. **Bottom-left panel:** seven mini-panels, implementing a number of plugins dedicated to the analysis of a specific selected region of the image.
4. **Bottom-right panel:** Flags Info, showing details on the flags already assigned to the image (see Section 11 for further details).

Besides the default navigation bar on the top of the image container, two other buttons are available to allow the user to open a new panel in a new window and to start the flagging procedure of the image (Section 11).

9 Report generation

One of main AIDA tools is the generation, handling and delivery of instrument data reports. In the AIDA framework, two different kinds of report are foreseen:

- **Periodic (Short/mid/long term data) report** : this is a pre-configured (through the compilation of a JSON configuration file) collection of monitored/analysed data, periodically produced by AIDA through automatic processes;
- **On demand report** : this is a collection of monitored/analysed data selected by any AIDA authenticated user, through the compilation of a JSON configuration file, produced on specific request.

All the generated reports are stored in the AidaHM local DB and accessible to all authenticated users.

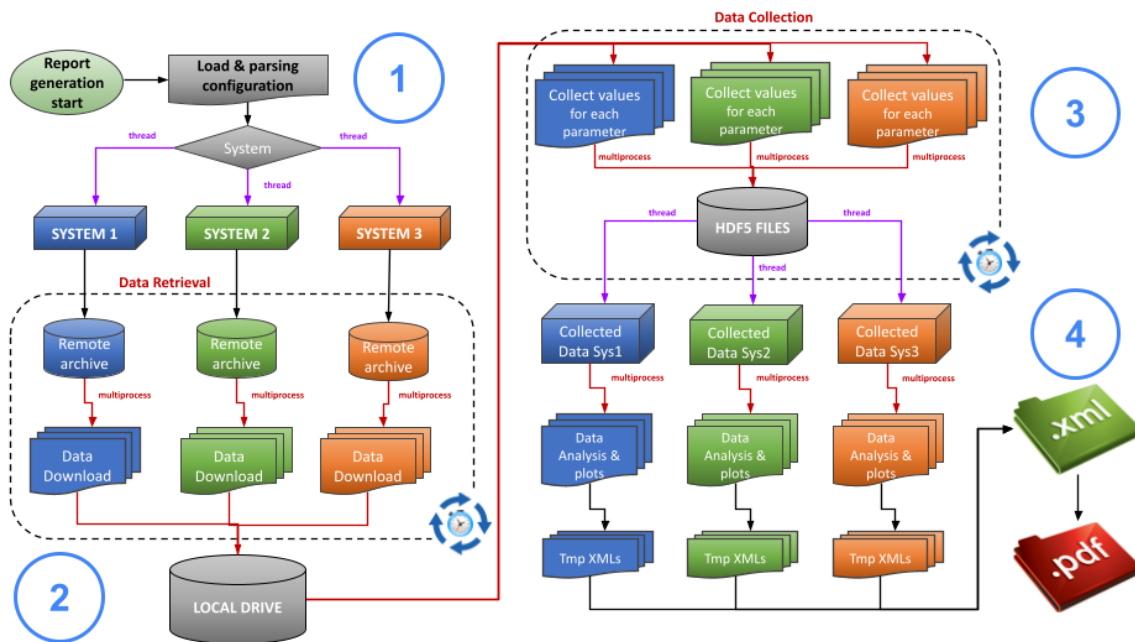


Figure 43: Reports generation flow: 1) The created configuration file is read by AIDA; 2) each system has a dedicated thread to retrieve metadata and files from the remote archives (with a pre-defined cadence to optimize performance) and store them on the AidaHM; 3) Data are extracted from metadata/files and stored in HDF5 files; 4) When all data have been stored for the related period, they are collected, elaborated and arranged into the XML and PDF outputs.

The common aspect among report types is the internal workflow mechanism, described in Figure 43, except for the execution timing (single execution chosen by the user in the case of on demand type, while periodically executed by an automatic process in the case of periodic type) and the setup performed by the user through the editing of a JSON configuration file.

Periodic reports

In the case of periodic reports, the AIDA system will periodically execute an autonomous process at a specified date and time, by collecting and organizing a predefined set of data, from any authorized data repository, under the form of a report document in PDF and XML formats. Such files will be then stored within the AidaHM filesystem and indexed in AIDA DB, depending on the operative conditions of the monitored system and the related context of AIDA execution. The setup and contents of periodic reports (systems, subsystems, parameters, time window and list of analysis tasks) are preliminary specified by any AIDA authenticated user, through a dedicated JSON configuration file, to be prepared via AIDA specific services.

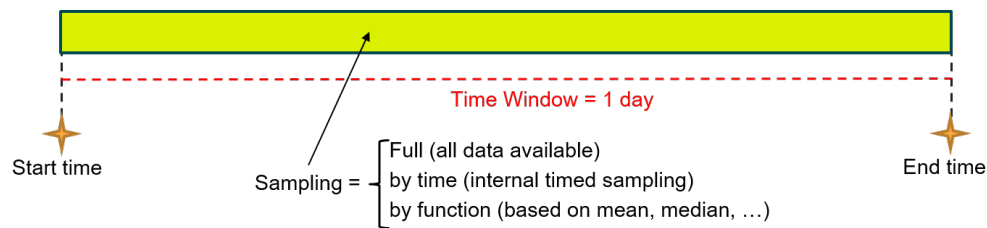


Figure 44: Schematic view of the data acquisition mode for periodic reports. The example is referred to a daily report. Yellow area indicates the time window valid for data acquisition. Sampling is related to the modes of data collection applicable within the Time Window, but for the current version of AIDA, only the *Full* option is available.

The Figure 44 shows a typical data retrieval/collection mode for the generation of a periodic report.

The information required to specify the range of reported data are provided by the user during the configuration file preparation:

- **Starting time** : date and time (UTC) from which to start the data retrieval and collection from the remote archive;

- **Time Window** : the duration (decimal hours) of the period in which existing data must be searched and collected. For a periodic report, this can be custom (user defined), daily, weekly or monthly;
- **Sampling mode** : the sampling to be used within the time window. This parameter was designed to define three different types of data sampling:
 - **Full**: all data physically acquired for the selected system available in the remote archive are collected and reported;
 - **By time**: the data collected and monitored are taken at selected regular time steps within the Time window;
 - **By function**: within the time window, the selection step is done using a cumulative function acting on all data taken between two successive steps. The function is chosen by user among mean (average), median, mode and other statistical functions made available;

NOTE: in the current version of AIDA, only the *Full* option is available.

On demand reports

In the case of on demand reports, any AIDA authenticated user is left free to prepare and generate an arbitrary number of customized report generations about any allowed system and its parameters, with any arbitrary date and time and related to any kind of data acquisition period. An on demand report execution, differently from the periodic type, can be executed only one-shot, i.e. not scheduled to be executed at regular time intervals. Figure 45 shows an example of data retrieval and collection mode for the generation of an on demand report.

The information required to specify the range of reported data is provided by the user during the configuration file preparation and includes some specific additional parameters, with respect to periodic type:

- **Starting time** : date and time (UTC) from which to start the data retrieval and collection from the remote archive;
- **Time Window** : the duration (decimal hours) of the period in which existing data must be searched and collected;

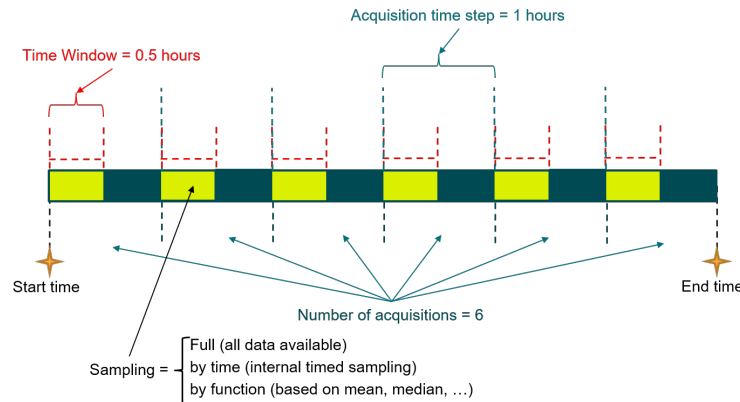


Figure 45: Schematic view of the data acquisition mode for on demand reports. The example is referred to a report generation composed by 6 consecutive acquisitions with a frequency of 1 hour (Acquisition time step) and a duration of each acquisition of 0.5 hours (Time Window). Yellow area indicates the time window valid for data acquisition, while dark green to the off-line time period.

- **Sampling mode** : the sampling to be used within the time window;
- **Number of acquisitions** : the total amount of acquisitions that must be repeated from starting date and time;
- **Acquisition Time Step** : the frequency (decimal hours) of the acquisition repetitions. Within each acquisition time step, the time window specifies the time percentage of the real data selection interval.

9.1 Report web application menu

Within the AIDA GUI, there are specific menu options dedicated to the setup and execution of report generation.

- Left Menu section “**Reports**”, four options are available:
 - “**New Configuration**”: used for the setup, editing and generation of the JSON configuration file (Figure 46). From this panel it is possible to edit and save configuration files, either as temporary (uncompleted) or completed, but also to execute the new report generation;
 - “**Generate Report**”: used to execute a report generation process, by selecting the related configuration file through any of existing files or by uploading a new one

- (Figure 47). Only reports based on completed configuration files can be generated (all configuration files stored and registered as uncompleted are not listed here);
- **“List Reports”**: used to list and download the reports already generated by all authenticated users (Figure 48). From this page the user can also flag a specific report by clicking on the related button (see Section 11 for further details on the flagging system);
 - **“Configuration Files”**: used to list various types of configuration files (including the uncompleted type) as compiled by all authenticated users (Figure 49).
- Left Menu section **“General Info”**: In order to simplify the editing of any report configuration file, the user can access to the list of systems, parameters and analysis operations that can be monitored and analyzed. Four options are available:
 - **“Systems Info”** (an example in Figure 50)
 - **“HKTМ Parameters”**: list of subsystems and related HKTМ parameters suitable for collection and analysis/monitoring operations;
 - **“Analysis Info”**
 - **“Available plots”**: list of available plot types within AIDA suitable for analysis on selected systems and parameters (Figure 51, left panel);
 - **“Available Statistics”**: list of available statistical estimators within AIDA suitable for analysis on selected systems and parameters (Figure 51, right panel);

9.2 Configuration of a report

Any kind of AIDA report requires the preliminary setup, through the editing of a JSON configuration file. This file will be stored in the AidaHM and indexed in the local DB. **The standard report setup procedure** to generate a new report configuration requires the following steps:

1. Select left menu option “New Configuration”
2. From the form “New Report Configuration” (as in Figure 46) the user must indicate the report type (periodic or on demand), starting date and time, time window, frequency, duration and sampling¹³ of data to be retrieved, downloaded and processed. This information will populate the header of the JSON file;

¹³In this case as well, only the *Full* option is available.

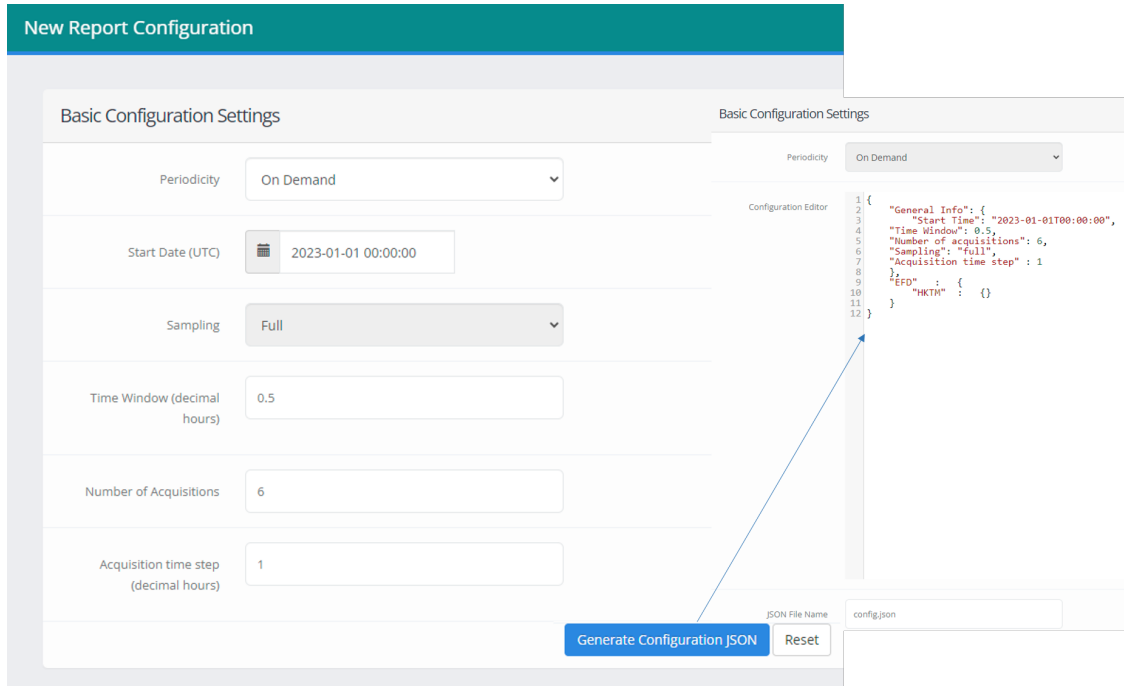


Figure 46: The configuration form activated by the Report menu option “New Configuration” and the internal editor (automatically opened by pressing the “Generate Configuration JSON” button) for the setup of a new report (both periodic or on demand).

3. After pressing the button “Generate Configuration JSON”, an editor panel will open, in which the user must list the system, subsystems and parameters to be collected, as well as the list of analysis operations to be applied to retrieved data and included in the report;
4. During the editing of the JSON file, the user can see the list of all allowed HKTM systems and parameters and the allowed analysis tools (plots/statistics) that can be used by AIDA, using the left menu “General Info” options;
5. Once the user has finished to edit the JSON file, he can perform three actions by clicking on the related button:
 - a. **“Store Temporary JSON” button** allows the user to suspend the editing and to resume it in a second time. In this case the temporary file is stored in the AidaHM and flagged into the local DB as “uncompleted”. The reports based on configuration files registered as uncompleted cannot be generated (no report generation processes can be launched).
 - b. **“Save Completed JSON” button** allows to store the final version of the configura-

Report Generation

Select Configuration File Origin

☒ Run from Existing Configuration File
 ☐ Upload New Configuration File

Generate Report from Existing Configuration File

10 records per page

Search:

Configuration File	Creation date	Last update	Owner	Recurrence	Start date	Operating Mode	
weekly.json	2024-04-19 13:22:43	2024-04-19 13:22:43		weekly	2023-01-01 12:00:00	NOMINAL	 
daily_long.json	2024-04-19 13:21:11	2024-04-19 13:21:11		daily	2023-01-01 04:00:00	NOMINAL	 
config_ondemand_long.json	2024-04-19 13:20:16	2024-04-19 13:20:16		ondemand	2023-01-01 04:00:00	NOMINAL	 
config2.json	2024-02-21 11:09:29	2024-02-21 11:09:29		ondemand	2023-01-01 04:00:00	NOMINAL	 
config.json	2024-02-08 14:18:28	2024-02-09 14:48:51		ondemand	2023-01-01 04:00:00	NOMINAL	 

Showing 1 to 5 of 5 entries

Previous

1

Next

Figure 47: The panel to execute a report generation task (opened by the Report menu option “Generate Report”). It is possible to execute an already configured report or a new one. An existing configuration can be used only if the system is in the same operating mode as indicated in the "Operating Mode" column.

Reports

Periodic Reports List

10 records per page

Search:

Report	Creation date	Owner	Recurrence	Start date	End date	Config file	Flag	Flag Creator	Flag Date	Flag Comments	
REP_20240419132452_13845-daily-20230112040000_20230112040000_BPD	2024-04-19 13:24:52		daily	2023-01-12 04:00:00	2023-01-12 04:00:00	daily_long.json			2024-04-23 08:19:05	-	 
REP_20240419132445_13840-daily-20230111040000_20230112040000_BPD	2024-04-19 13:24:45		daily	2023-01-11 04:00:00	2023-01-12 04:00:00	daily_long.json			-	-	 
REP_20240419132432_13840-daily-20230110040000_20230111040000_BPD	2024-04-19 13:24:32		daily	2023-01-10 04:00:00	2023-01-11 04:00:00	daily_long.json			-	-	 
REP_20240419132424_13840-daily-20230109040000_20230110040000_BPD	2024-04-19 13:24:24		daily	2023-01-09 04:00:00	2023-01-10 04:00:00	daily_long.json			2024-04-19 13:30:34	All OK	 
REP_20240419132305_13819-weekly-20230101120000_20230108120000_BPD	2024-04-19 13:23:05		weekly	2023-01-01 12:00:00	2023-01-08 12:00:00	weekly.json			2024-04-19 13:31:12	Warning comment	 

Showing 1 to 5 of 5 entries

Previous

1

Next

On Demand Reports List

10 records per page

Search:

Report	Creation date	Owner	Start date	End date	Config file	Flag	Flag Creator	Flag Date	Flag Comments	
REP_20240419132415_13841-ondemand-20230101040000_20230121040000_BPD	2024-04-19 13:24:15		2023-01-01 04:00:00	2023-01-21 04:00:00	config_ondemand_long.json			2024-04-19 13:32:50	Error!	 
REP_202402211112540_13833-ondemand-20230101040000_20230109040000_BPD	2024-02-21 11:25:40		2023-01-01 04:00:00	2023-01-09 04:00:00	config2.json			-	-	 
REP_20240209130554_13797-ondemand-20230101040000_20230109040000_BPD	2024-02-09 13:05:44		2023-01-01 04:00:00	2023-01-01 09:00:00	config.json			-	-	 

Showing 1 to 3 of 3 entries

Previous

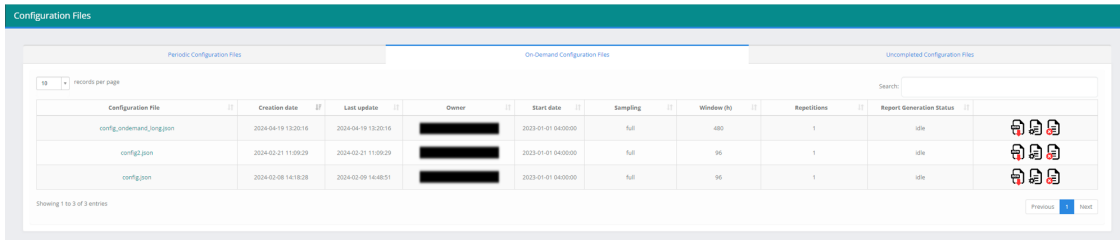
1

Next

Figure 48: Example of report list panel (opened by the Report menu option “List Reports”).

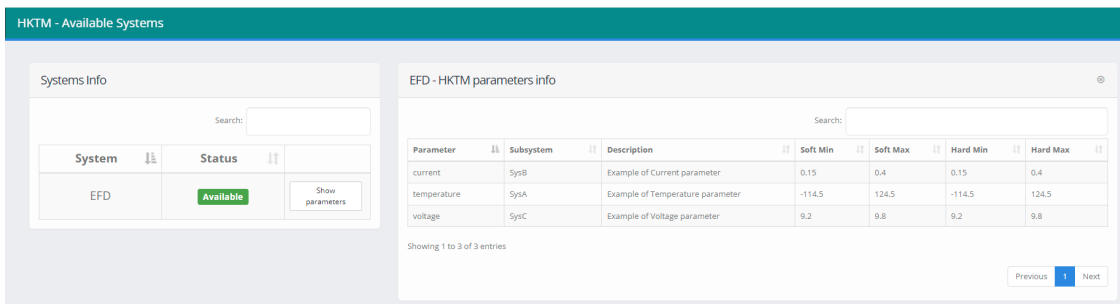
tion file into the AidaHM, indexed in the local DB, flagged as “completed”, without launching the report generation process. The report generation can be started at any time by accessing to the “Generate Report” page (Figure 47);

- “Generate Report” button** allows to store the configuration file into the AidaHM and launch the process in charge to retrieve remote data and generate the report;



Configuration File	Creation date	Last update	Owner	Start date	Sampling	Window (s)	Repetitions	Report Generation Status
config_demand_long.json	2024-04-19 13:20:16	2024-04-19 13:20:16	[Redacted]	2023-01-01 04:00:00	full	480	1	idle
config.json	2024-02-21 11:09:29	2024-02-21 11:09:29	[Redacted]	2023-01-01 04:00:00	full	96	1	idle
config.json	2024-02-08 14:18:28	2024-02-09 14:48:51	[Redacted]	2023-01-01 04:00:00	full	96	1	idle

Figure 49: The panel opened by the Report menu option “Configuration Files”. It shows the list of all existing report configuration files (with the status of associated report generation task, e.g. idle/scheduled/running), either periodic, on demand and uncompleted, the latter referred to configuration files partially edited but still to be completed. Only completed and idle configuration files can be launched. Each user can edit or delete his/her configuration files while administrator(s) can edit/delete every file.



Parameter	Subsystem	Description	Soft Min	Soft Max	Hard Min	Hard Max
current	SysB	Example of Current parameter	0.15	0.4	0.15	0.4
temperature	SysA	Example of Temperature parameter	-114.5	124.5	-114.5	124.5
voltage	SysC	Example of Voltage parameter	9.2	9.8	9.2	9.8

Figure 50: Example of "System Info" table, reporting la list of parameters suitable for collection and analysis/monitoring operations.

The configuration files, as representatives of related reports (any report is always associated with a configuration file), can be monitored in terms of report generation process status. In such terms, a report generation job can be in **idle** status if not running, **scheduled** if launched but still not running, or **running** if under execution. The scheduled status is always referred to a report generation process in which the starting time, indicated in the associated configuration file, is projected into the future. This is particularly frequent in case of the generation of periodic reports.

The list of all configuration files, generated by all authenticated users, their report generation status (idle/scheduled/running) and other related metadata can be accessed through specific panels, divided into three tabs (periodic/ondemand/uncompleted) (see Figure 49).

Once a report is generated, an e-mail will be sent to the user to notify the report creation, with a link to directly access to its XML and PDF version (Figure 52).

All data included in the configuration file will be collected and organized in the report in the most readable way, by subdividing data based on the specific reference instrument and sub-system. As showed in the bottom-left panel of Figure 53, each parameter is listed with its status, organized as a traffic light, in which a green status indicates a normal condition, yellow a warning and red a failure or severe anomaly detected. The status is assigned based on the comparison between the assumed values and any anomaly conditions supplied with the parameter or as the result of any correlation/thresholding operation as specified by the user

Available Plots

Plots Info

Search:

Name	Report config name	Parameters
Histogram	histogram	Bin Size, Number of Bins
Scatter Plot	scatter	X
Trend Analysis	trend	

Available Statistics

Statistics Info

Search:

Name	Report config name	Parameters
Biweight_Mean	biweight mean	iterMax, epsilon
Kurtosis	kurtosis	
MAD	mad	
Max	max	
Mean	mean	
Median	median	
Min	min	
Mode	mode	precision
NMAD	nmad	
Percentile	percentile	quantile, interpolation
RMS	rms	
Sigma_Clip	sigma clip	sigma, function
Skewness	skewness	
Standard_Deviation	standard deviation	
Variance	variance	

Figure 51: "Available Plots" (**left panel**) and "Available Statistics" (**right panel**) tables from "General Info" section. In each row, item name, key to use to tag an operation and setting keys to configure each operation in a report configuration file.

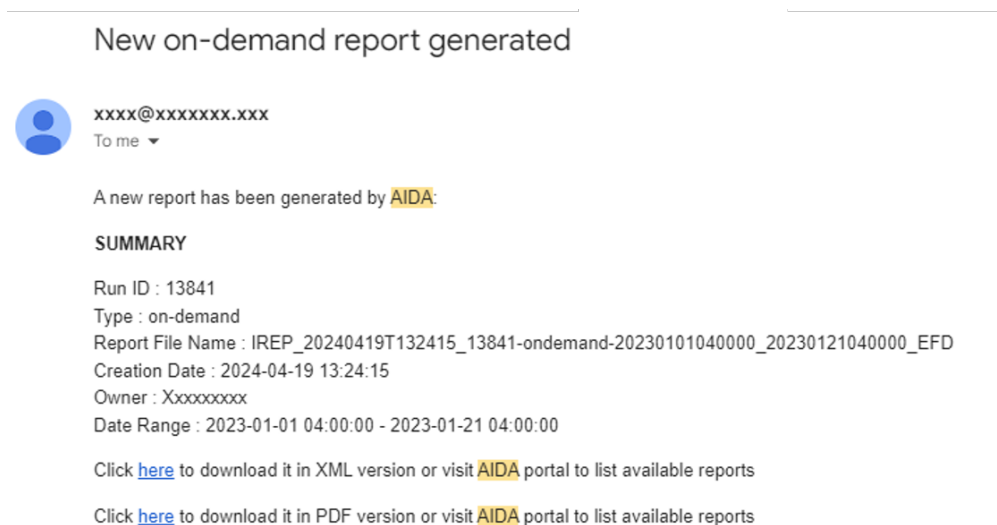


Figure 52: Example of new generated report notification email.

within the configuration file.

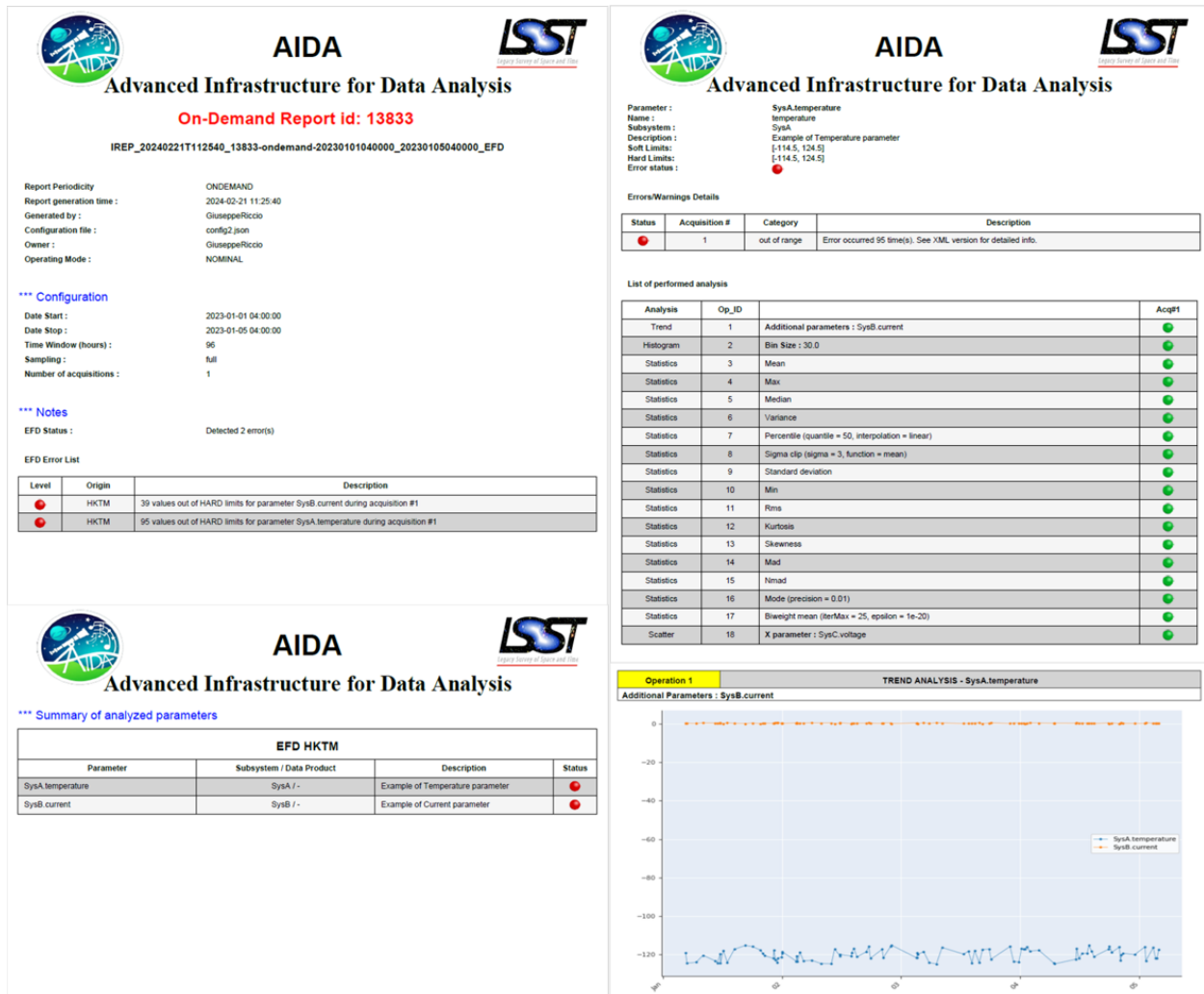


Figure 53: Example of an on demand report PDF output file (same layout for periodic reports).

9.3 Reports JSON configuration files editing guide

In a report configuration file all the info required to configure the experiment is collected as a JSON data format, basically information is nested along a dedicated branch. The main structure of a report JSON configuration file is common for each use case but, due to the wide range of DPs, subsystems, parameters and options, the single branch can strongly vary from case to case. The general structure can be divided into two parts: i) a **General Info** section, storing some general experiment settings; ii) one or more **System branches**, section specifically structured to configure experiments on each system. Even if the single *System*

branch has a unique structure depending on the system to configure, all the branches have a general common structure:

- **First key:** name of the system ("EFD");
- **Second key:** name of data origin ("HKTM");
- **Inner settings:** a number of nested pairs (key-value) collecting all the required info to characterize the parameters to monitor;
- **Operations:** for each configured parameter, the list and settings of operations to perform on it.

A complete list of parameters and analysis tools available for configuration file creation is reported in the pages linked in "General Info" section of the sidebar (Figures 50 and 51).

Below is a more in-depth description of the configuration files.

9.3.1 General Info

As already mentioned in Section 9.2, some general settings must be defined and stored under the "General Info" branch for every configuration file:

- **Start time:** date and time (UTC) from which to start the data retrieval and collection from the remote archive;
- **Time Window:** the duration (decimal hours) of the period in which existing data must be searched and collected;
- **Number of acquisitions:** (Required only for on demand reports) the total amount of acquisitions that must be repeated from starting date and time;
- **Acquisition time step:** (Required only for on demand reports if *Number of acquisitions* > 1): the frequency (decimal hours) of the acquisition repetitions;
- **Sampling:** the sampling to be used within the time window (For the current version of AIDA, only value "full" is allowed);

```
"General Info": {
  "Start Time": "2023-01-01T04:00:00",
  "Time Window": 24,
  "Number of acquisitions": 1,
  "Sampling": "full"
}
```

9.3.2 Inner settings

The central part of a use case (System-Data origin) branch is devoted to the definition of the parameters to monitor. It strongly depends on the related use case, but tries to follow the part related to parameter configuration in the analogous form for plot generation (see Section 5 for further details), except for few filters in the form, used to simplify its filling.

For EFD data the structure of the inner part is the following:

```
"EFD": {
  "HKTM": {
    "<subsystem#1>" : {
      "<field#1.parameter#1>" : {
        "Operation_1": {
          <operation info. See dedicated section.>
        },
        ...
        "Operation_<N>": {
          <operation info. See dedicated section.>
        }
      },
      ...

      "<field#N.parameter#1>": {
        "Operation_1": {
          <operation info. See dedicated section.>
        },
        ...
      }
    }
  }
}
```

```

        "Operation_<N>": {
            <operation info. See dedicated section.>
        }
    },
    ...
},
...

"<subsystem#n>" : {
    ...
}
}
}

```

For FAKE data the structure of the inner part is the following:

```

"FAKE": {
    "HKTM": {
        "<subsystem#1>" : {
            "<parameter#1>" : {
                "Operation_1": {
                    <operation info. See dedicated section.>
                },
                ...
                "Operation_<N>": {
                    <operation info. See dedicated section.>
                }
            },
            ...

            "<parameter#M>": {
                "Operation_1": {
                    <operation info. See dedicated section.>
                }
            }
        }
    }
}

```

```

    },
    ...
    "Operation_<N>": {
        <operation info. See dedicated section.>
    }
}
},
...

"<subsystem#n>" : {
    ...
}
}
}

```

NOTE: in each branch, at the same nested level, more then one configuration tag can be used (for example, one or more <Parameter>, <Data product>, ...), obviously following the JSON typing rules.

9.3.3 Operations

For each listed parameter, it is needed to define the operations to perform. They must listed by using the tag "Operation_<#n>" as key, where <#n> is a progressive number starting from 1. For each "Operation_<#n>" key, the related value will be a nested dictionary containing all the info needed to set up the operation itself and summarized in Table 1.

The lists of all available plots and statistics and their tags and settings are shown on the pages linked in *General Info* → *Analysis Info* (see Figure 51)

Key	Required/Option	Allowed values	Notes
"Type"	Always required	"trend"	"Additional parameters" applicable
		"scatter"	"Additional parameters" applicable "X" required
		"histogram"	"Additional Parameters" applicable "Bin Size" or "Number of Bins" required
		"biweight mean"	"Parameters" required
		"kurtosis"	-
		"mad"	-
		"max"	-
		"mean"	-
		"median"	-
		"min"	-
		"mode"	"Parameters" required
		"nmad"	-
		"percentile"	"Parameters" required
		"rms"	-
		"sigma clip"	"Parameters" required
		"skewness"	-
		"standard deviation"	-
		"variance"	-
"Additional Parameters"	plots only - optional	see Section 9.3.4	-
"X"	"scatter" only - required	see Section 9.3.4	-
"Bin Size"	"histogram" only - required (see notes)	see Section 9.3.4	alternative to "Number of Bins"

Key	Required/Option	Allowed values	Notes
"Number of Bins"	"histogram" only - required (see notes)	see Section 9.3.4	alternative to "Bin Size"
"Parameters"	required for some statistics	see Section 9.3.4	-

Table 1: List of allowed keys for operation configuration.

```

"Operation_1" : {
    "Type" : "trend",
    "Additional Parameters" : <See dedicated section>,
},
"Operation_2" : {
    "Type" : "scatter",
    "X" : <See dedicated section.>,
    "Additional Parameters" : <See dedicated section>
},
"Operation_3" : {
    "Type" : "histogram",
    "Bin Size" : <See dedicated section.>,
    "Additional Parameters" : <See dedicated section>
},
"Operation_4" : {
    "Type" : "histogram",
    "Number of Bins" : <See dedicated section>,
    "Additional Parameters" : <See dedicated section>
},
"Operation_5" : {
    "Type" : <statistics WITHOUT additional settings>
},
...
"Operation_<N>" : {
    "Type" : <statistics WITH additional settings>,
    "Parameters" : {
        <See dedicated section>
    }
}

```

```
}
}
```

9.3.4 Additional Settings

Some operations requires other settings to be defined. In particular:

- All the plots ("trend", "scatter", "histogram") give the possibility to display more than one parameter on the same plot or on adjacent subplots, by defining the **"Additional Parameters"** item;
- "scatter" plot requires that the parameter on x axis to be defined ("**X**" item);
- "histogram" plot requires to define the size of the bins ("**Bin Size**" item) or the number of bins to populate ("**Number of Bins**");
- some statistics (namely "biweight mean", "mode", "percentile" and "sigma clip") requires other settings to be defined, by adding the **"Parameters"** dictionary.

Additional Parameters

In order to plot other parameters together with the one under monitoring, the user has to use the **"Additional Parameters"** key under the "Operation_<#N>" section. The value associated to the key is a list of all the parameters to use (see below). All parameters will be arranged in a unique plot and/or adjacent subplots, depending on its type.

```
"Operation_1" : {
  "Type" : "trend",
  "Additional Parameters" : [<add_param #1>, <add_param #2>, ... <
    add_param #N>]
}
```

An *additional parameter* string must contain all the info characterizing the parameter, and depends on the use case, similarly to the main parameter. For EFD data, the structure for an additional parameter is:

`<subsystem>.<field>.<parameter>`

while, for FAKE data:

`<subsystem>.<parameter>`

Scatter x axis

The definition of the parameter to use on the scatter plot x axis is made by the **<X>** key and a string as value. The structure of the string is the same as for an additional parameter.

```
"Operation_1" : {
  "Type" : "scatter",
  "X" : "<X parameter>"
}
```

Bins characterization

The binning is required to configure a histogram operation. The user can choose between two different tags:

- **"Bin Size"** : width of bins. Its format is *float*;
- **"Number of Bins"** : the number of bins to use (format *int*);

```
"Operation_1" : {
  "Type" : "histogram",
  "Bin Size" : 0.05
},
"Operation_2" : {
  "Type" : "histogram",
  "Number of Bins" : 15
}
```

Statistical Parameters

There are some statistical indicators that require to be configured by using additional settings. This settings are collected into a dictionary associated to the "**Parameters**" tag under the "Operation_<#N>" section: the name of the setting will assume the role of key of the dictionary.

All required extra settings are listed on the page linked to *General Info* → *Analysis Info* → *Available Statistics* (Figure 51) and summarized in Table 2.

Statistics Tag	Extra settings	Format	Required/optional	Description
"biweight mean"	"iterMax"	int	optional (default 25)	Maximum number of iterations to approximate the standard deviation of the distribution
	"epsilon"	float	optional (default 1e-20)	Minimum value of the standard deviation to continue iterations
"mode"	"precision"	float	optional (default 0)	Precision for mode evaluation. If precision is greater than 0, data will be arranged into bins of width = 2*precision and the median of values in each bin will be considered for calculation. Otherwise the single values will be considered.
"percentile"	"quantile"	float in [0,100]	required	Percentile to compute, which must be between 0 and 100 inclusive
	"interpolation"	"linear" "lower" "higher" "nearest" "mid-point"	required	Interpolation method to use when the desired quantile lies between two data points $i < j$
"sigma clip"	"sigma"	float	required	The number of standard deviations to use for both the lower and upper clipping limit

	"function"	"biweight" "mad" "median" "mean" "nmad"	required	The statistics used to compute the center value for the clipping
--	------------	---	----------	--

Table 2: List of statistical indicators which require extra settings.

9.4 Some Examples

In this section, two examples of configuration file created for EFD and FAKE data are shown, compiled by following the rules described in Section 9.3.

9.4.1 EFD data example

```
{
  "General Info": {
    "Start Time": "2026-05-02T00:00:00",
    "Time Window": 24,
    "Number of acquisitions": 1,
    "Sampling": "full"
  },
  "EFD": {
    "HKTM": {
      "MTM2": {
        "temperature.ring0": {
          "Operation_1": {
            "Type": "trend",
            "Additional Parameters": ["MTM2.temperature.ring1", "MTM2
              .temperature.ring2"]
          },
          "Operation_2": {
            "Type": "scatter",
            "X": "MTM2.temperature.ring1"
          }
        }
      }
    }
  },
}
```

```

"tangentForce.applied3": {
  "Operation_1": {
    "Type": "histogram",
    "Number of Bins": 4
  }
},
"temperature.ring1": {
  "Operation_1": {
    "Type": "mean"
  },
  "Operation_2":{
    "Type" : "max"
  },
  "Operation_3":{
    "Type" : "median"
  },
  "Operation_4": {
    "Type": "variance"
  },
  "Operation_5": {
    "Type": "percentile",
    "Parameters" : {"quantile" : 50, "interpolation":"
    linear"}
  },
  "Operation_6": {
    "Type": "sigma clip",
    "Parameters" : {"sigma" : 3, "function":"mean"}
  },
  "Operation_7": {
    "Type": "standard deviation"
  },
  "Operation_8": {
    "Type": "min"
  },
  "Operation_9": {
    "Type": "rms"
  }
}

```

```

    },
    "Operation_10": {
      "Type": "kurtosis"
    },
    "Operation_11": {
      "Type": "skewness"
    },
    "Operation_12": {
      "Type": "mad"
    },
    "Operation_13": {
      "Type": "nmad"
    },
    "Operation_14": {
      "Type": "mode",
      "Parameters" : {"precision":0.01}
    },
    "Operation_15": {
      "Type": "biweight mean",
      "Parameters" : {"iterMax" : 25, "epsilon":"1e-20"}
    }
  }
}
}
}
}

```

9.4.2 FAKE data example

```

{
  "General Info": {
    "Start Time": "2023-01-01T04:00:00",
    "Time Window": 24,
    "Number of acquisitions": 1,
    "Sampling": "full"
  },

```

```

"EFD"      :   {
  "HKTM"    :   {
    "SysA":{
      "temperature":{
        "Operation_1" : {
          "Type" : "trend",
          "Additional Parameters" : ["SysB.current"]
        },
        "Operation_2" : {
          "Type" : "histogram",
          "Bin Size" : 30
        },
        "Operation_3":{
          "Type" : "mean"
        },
        "Operation_4":{
          "Type" : "max"
        },
        "Operation_5":{
          "Type" : "median"
        },
        "Operation_6": {
          "Type": "variance"
        },
        "Operation_7": {
          "Type": "percentile",
          "Parameters" : {"quantile" : 50, "interpolation": "linear"}
        },
        "Operation_8": {
          "Type": "sigma clip",
          "Parameters" : {"sigma" : 3, "function": "mean"}
        },
        "Operation_9": {
          "Type": "standard deviation"
        },
      },
    },
  },

```

```

        "Operation_10": {
          "Type": "min"
        },
        "Operation_11": {
          "Type": "rms"
        },
        "Operation_12": {
          "Type": "kurtosis"
        },
        "Operation_13": {
          "Type": "skewness"
        },
        "Operation_14": {
          "Type": "mad"
        },
        "Operation_15": {
          "Type": "nmad"
        },
        "Operation_16": {
          "Type": "mode",
          "Parameters" : {"precision":0.01}
        },
        "Operation_17": {
          "Type": "biweight mean",
          "Parameters" : {"iterMax" : 25, "epsilon":"1e-20"}
        },
        "Operation_18": {
          "Type": "scatter",
          "X" : "SysC.voltage"
        }
      }
    },
    "SysB":{
      "current":{

```

```
    "Operation_1" : {  
      "Type" : "trend"  
    },  
    "Operation_2" : {  
      "Type" : "histogram",  
      "Number of Bins" : 3  
    }  
  }  
}  
  
}  
  
}
```

10 Analyze local data

AIDA allows to generate plots and statistics from FITS table or CSV files from local machine. The user can upload a new local file (Figure 54, left panel) or select an already existing file, previously uploaded during the current user session (Figure 54, right panel), by clicking "Analyze Local Data" from left menu.

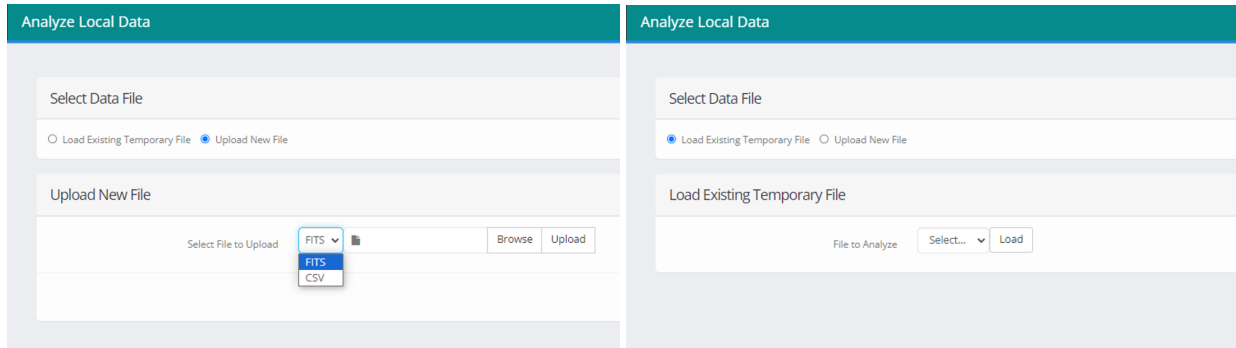


Figure 54: File selection form for "Analyze Local Data" usecase.

Once the file is selected/uploaded, the user can configure the experiment by defining the analysis type, parameter(s) to show, statistics to perform and all the required info for the selected analysis (see Section 5). Fields for parameter configuration are automatically extracted from the selected/uploaded file.

An example of form for trend analysis is shown in Figure 55.

Analyze Local Data

Configure Analysis

Uploaded File

testfile.fits

Success

Analysis to perform

Scatter plot

X axis parameter

kron

Y axis parameters

mag

Add more...

Advanced Statistics

Submit

Reset

Figure 55: Example of configuration for local file plot generation. Fields for parameter configuration are automatically extracted from the selected/uploaded file.

11 Flagging system

It is possible to associate a semaphore-like flag to all AIDA products, except for plots generated from local data, by clicking on the related button, located as follows:

- **"Plots"**: Disk icon on the plot navigation bar (Figure 34);
- **"Statistics"**: "Save Statistics" button (Section 6);
- **"Images"**: "Flag image" button (Section 8);
- **"Reports"**: related "Add flag" button in reports lists (Figure 48).

On click, a modal window will open, specific for each product (Figure 56).

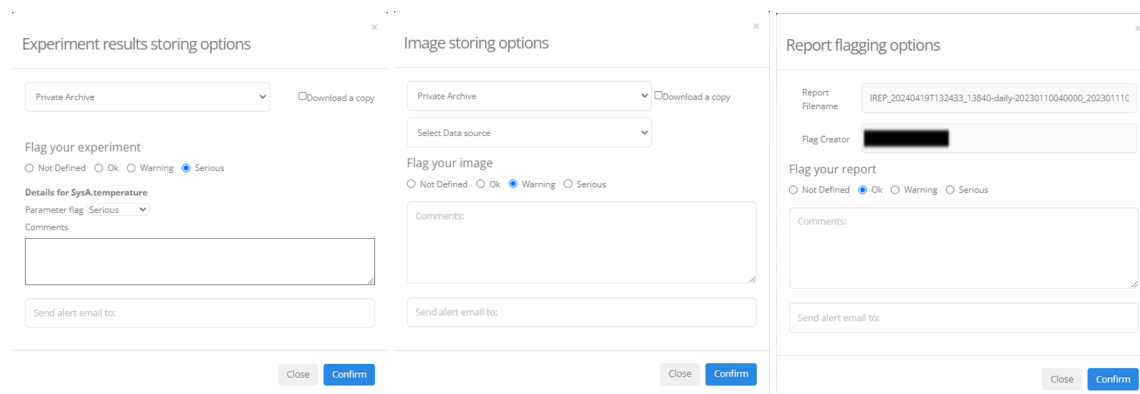


Figure 56: Flags modal window for different products: plots and statistics (**left**), images (**center**), reports (**right**).

Depending on the product, the user can fill the displayed form with some general info (for example if the stored experiment is public or personal) while some other fields are pre-compiled by AIDA (for example the report filename). A copy of the PDF experiment report can be downloaded too, except for reports. Then, it is possible to set the flag, add some comments describing the assigned flag and send an alert e-mail to a potentially interested person (Figure 57). If the product is a plot or a statistical analysis (Figure 56, left panel), it is possible to add a flag and related comment for each involved parameter.

Report flagged



xxxx@xxxxxxxx.xxx
To me ▼

A new report has been flagged in AIDA repository by user User_1.

For additional info, please contact User_1 directly.

Report Name : IREP_20240419T132443_13840-daily-20230111040000_20230112040000_EFD

Status: OK

Comments : Report flag test

Figure 57: Example of flag alert email for reports.

12 Logging system

This is a section dedicated to describe the logging system designed and developed to inform users about the internal status of AIDA and interaction results. The logging system is strictly connected with the handling of all errors generated by AIDA, either in terms of errors due to users and due to internal module operations. At a higher level, since AIDA is a web application, the logging system is based on the message system as provided by Apache. AIDA is provided with a logging system to report different information about its activity. In the current version, three different kinds of logs are available:

1. Reports logging: for each report generation pipeline, a log file is stored, identified by the report runID and reporting the status and the duration of each operation running step. In the log file the tracebacks of eventual run anomalous events (warnings/errors) are also reported, while specific errors about data acquisition and analysis are reported in the generated reports themselves. These logs are available in `<root>/scripts/logs` directory;
2. History: the history of the user actions is stored into the AIDA local DB, as a stack with a limit per user defined by administrator(s) (see Section 4.2.6). Reported actions include:
 - a Login/Logout;
 - b Plot/Statistics/ML experiment status;
 - c Creation of configuration files (both temporary and complete);
 - d Update of configuration files;
 - e Report generation from just a created, already existing or uploaded configuration file;
 - f Pause/Kill/Resume of a report generation (only for administrators);
 - g Operations on images from Image Explorer;
 - h Experiment flagging and storing into AIDA DB.
3. Web App errors: all the errors/warnings generated by the AIDA web application malfunctioning are reported into two kinds of file:
 - a Apache error.log file, for all the errors concerning the server engine;
 - b HTML files into the directory `scripts/cgi-scripts` for all CGI errors related to the client-server interaction with AJAX and Python.

In addition to the logging system, the user action history is displayed on the “History” tab of the user dashboard (Section 4.2.2).

13 Error handling system

In terms of error handling, the strategy is to distinguish between warnings and errors. The following tables report the error handling lists, divided by functionality.

13.1 Web Application Navigation

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-WAN-01	login	user try to login	user not registered	message	Error Logging In! Please retry	Redirect to index	alert box
E-WAN-02	login	user try to login	bad credentials	message	Error Logging In! Please retry	Redirect to index	alert box
E-WAN-03	signup	user try to register	user already exists	message	Registration failed! User already present in AIDA DB. Please, change name and/or e-mail address or contact AIDA admin.		alert box
E-WAN-04	signup, installation	user compiles registration form	not valid email address	error text appears	Please enter a valid email address.		form
E-WAN-05	signup, installation	user compiles registration form	password without a digit or/and an uppercase letter	error text appears	An upper case letter and a digit required		form
E-WAN-06	signup, installation	user compiles registration form	password with length < 8 characters	error text appears	At least 8 characters required		form
E-WAN-07	signup, installation	user compiles registration form	wrong password confirmation	error text appears	Please enter the same value again		form
E-WAN-08	signup, installation	user compiles registration form	wrong captcha	error text appears	Registration failed! Incorrect CAPTCHA		form
E-WAN-09	signup, installation	user try to register	unable to connect to local DB	message	Registration failed! Impossible to connect to database. Please contact AIDA admin	Redirect to index	alert box

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-WAN-10	dashboard view - tab "Reports"	try to show "Available Reports" tree	unable to connect to local DB	error text appears	Unable to load directory tree. Please retry or contact AIDA admin		text in related div
E-WAN-11	dashboard view - tab "Reports"	try to show "Configuration Files" tree	unable to connect to local DB	error text appears	Unable to load directory tree. Please retry or contact AIDA admin		text in related div
E-WAN-12	dashboard view - tab "Reports"	try to show "Running Reports Generation" table	unable to connect to local DB	error text appears	Unable to retrieve Running Reports data. Please retry or contact AIDA admin		text in related div
E-WAN-15	dashboard view - tab "History"	try to show "User" history tree	unable to connect to local DB	error text appears	Unable to retrieve History data. Please retry or contact AIDA admin.		text in related div
E-WAN-16	dashboard view - tab "History"	try to show "Global" history tree	unable to connect to local DB	error text appears	Unable to retrieve History data. Please retry or contact AIDA admin.		text in related div
E-WAN-17	dashboard view - tab "Stored Experiments"	try to show "Public Archive" table by parameter	unable to connect to local DB	error text appears	Unable to retrieve data. Please retry or contact AIDA admin.		text in related div
E-WAN-18	dashboard view - tab "Stored Experiments"	try to show "Personal Archive" table by parameter	unable to connect to local DB	error text appears	Unable to retrieve data. Please retry or contact AIDA admin.		text in related div
E-WAN-19	dashboard view - tab "Stored Experiments"	try to show "Public Archive" table by experiment	unable to connect to local DB	error text appears	Unable to retrieve data. Please retry or contact AIDA admin.		text in related div
E-WAN-20	dashboard view - tab "Stored Experiments"	try to show "Personal Archive" table by Experiment	unable to connect to local DB	error text appears	Unable to retrieve data. Please retry or contact AIDA admin.		text in related div

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-WAN-21	dashboard view - tab "Administration - Users"	try to show "Pending Requests" table	unable to connect to local DB	error text appears	Unable to retrieve data. Please retry or contact AIDA admin.		text in related div
E-WAN-22	dashboard view - tab "Administration - Users"	try to show "Active Users" table	unable to connect to local DB	error text appears	Unable to retrieve data. Please retry or contact AIDA admin.		text in related div
E-WAN-23	dashboard view - tab "Administration - Users"	try to show "Deactivated Users" table	unable to connect to local DB	error text appears	Unable to retrieve data. Please retry or contact AIDA admin.		text in related div
E-WAN-24	"View Local Data"	try to show "User Files" table	unable to connect to local DB	error text appears	Unable to retrieve data. Please retry or contact AIDA admin.		text in related div
E-WAN-25	"View Local Data"	try to show "Global Files" table	unable to connect to local DB	error text appears	Unable to retrieve data. Please retry or contact AIDA admin.		text in related div
E-WAN-26	"List Reports"	try to show "Periodic Report List" table	unable to connect to local DB	error text appears	Unable to retrieve Reports data from local DB. Please retry or contact AIDA admin.		text in related div
E-WAN-27	"List Reports"	try to show "On-Demand Report List" table	unable to connect to local DB	error text appears	Unable to retrieve Reports data from local DB. Please retry or contact AIDA admin.		text in related div
E-WAN-28	"Configuration Files"	try to show "Periodic Configuration Files" table	unable to connect to local DB	error text appears	Unable to retrieve Configuration Files data from local DB. Please retry or contact AIDA admin.		text in related div

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-WAN-29	"Configuration Files"	try to show "On-Demand Configuration Files" table	unable to connect to local DB	error text appears	Unable to retrieve Configuration Files data from local DB. Please retry or contact AIDA admin.		text in related div
E-WAN-30	"Configuration Files"	try to show "Uncompleted Configuration Files" table	unable to connect to local DB	error text appears	Unable to retrieve Configuration Files data from local DB. Please retry or contact AIDA admin.		text in related div
E-WAN-31	"Generate Report"	try to show "Generate Report from Existing Configuration File" table	unable to connect to local DB	error text appears	Unable to retrieve data. Please retry or contact AIDA admin.		text in related div
E-WAN-32	dashboard view - tab "Administration - Users"	administrator inserts wrong password on changing General Settings, Edit Remote Repositories, Edit STMP, Ingestion Settings	wrong password confirmation	message	ERROR! Invalid password		alert box
E-WAN-33	forgot password	user try to reset their password	user not registered	message	ERROR! No user is registered with this email address!	Redirect to index	alert box
E-WAN-34	forgot password	user try to reset their password	user not active	message	ERROR! User is not active	Redirect to index	alert box

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-WAN-35	forgot password	user try to reset their password	unable to connect to local DB	message	ERROR! Impossible to connect to DB. Please retry in few minutes or contact AIDA Admin!	Redirect to index	alert box
E-WAN-36	forgot password	user try to reset their password	unable to send recovery link by email	message	ERROR! Impossible to send password recovery link. Please, contact AIDA Admin	Redirect to index	alert box
E-WAN-37	forgot password	user try to reset their password	unable to complete request	message	ERROR! Impossible to complete your request. Please, try later or contact AIDA Admin	Redirect to index	alert box
E-WAN-38	password reset	AIDA try to reset user password	unable to connect to local DB	message	ERROR! Impossible to connect to local DB. Please retry or contact AIDA admin.	Redirect to index	alert box
E-WAN-39	dashboard view - tab "Administration - Settings"	administrator try to change operating mode	unable to update	message	ERROR! Impossible to update settings. Please, retry later or contact AIDA Admin.		alert box
E-WAN-40	dashboard view - tab "Administration - Settings"	administrator tries to Export data	unable to export data	message	ERROR! Impossible to create backup file. Please, retry later or contact AIDA Admin		alert box
E-WAN-41	dashboard view - tab "Administration - Settings"	administrator tries to enable/disable system	unable to update settings	message	ERROR! Impossible to update systems enabled. Please retry later or contact AIDA admin		alert box
E-WAN-42	dashboard view - tab "Administration - Settings"	administrator tries to update general settings	unable to update settings	message	Error updating settings! <error description>		alert box

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-WAN-43	dashboard view - tab "Administration - Settings"	administrator tries to update SMTP settings	unable to update settings	message	Error updating SMTP settings! Impossible to store SMTP configuration file		alert box
E-WAN-44	dashboard view - tab "Administration - Settings"	administrator tries to update ingestion settings	unable to update settings	message	Error updating SMTP settings! Impossible to store Ingestion configuration file		alert box
E-WAN-45	dashboard view - tab "Administration - Settings"	administrator tries to update system settings	unable to update settings	message	Error! Impossible to update configuration file		alert box
E-WAN-46	dashboard view - tab "Administration - Users"	administrator tries to activate/deactivate/remove user	unable to update user status	message	Error! Impossible to complete the requested operation		alert box
E-WAN-47	dashboard view - tab "Administration - Settings"	administor changes settings in "General Settings" and submits	a field is not filled	error text appears	This field is required.		form
E-WAN-48	dashboard view - tab "Administration - Settings"	administor changes "Communication E-mail" in "General Settings" and submits	wrong e-mail(s) format	error text appears	Please, enter valid email addresses separated by commas		form

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-WAN-49	dashboard view - tab "Administration - Settings"	administrators changes "History records per user" in "General Settings"	inserted number less than 0	error text appears	Please enter only digits.		form
E-WAN-50	dashboard view - tab "Administration - Settings"	administrators changes "Processors per system" in "General Settings"	inserted number less than 1	error text appears	Please enter a value greater than or equal to 1.		form

Table 3: Web Application Navigation errors

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
I-WAN-01	dashboard view - tab "Administration - Settings"	administrator try to change operating mode	submission without changing operating mode	message	INFO: Operating mode unchanged. Nothing to do.		alert box
I-WAN-02	dashboard view - tab "Administration - Settings"	user try to Export data	impossible to store operation to history	message	impossible to store operation to history		alert box
I-WAN-03	dashboard view - tab "Administration - Settings"	user try to Export data without selecting any items	no item selected	message	Nothing to backup. Please, select at least one item to backup		alert box

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
I-WAN-04	dashboard view - tab "Administration - Settings"	administrator tries to update system settings without changing any items	no item selected	message	INFO: Systems unchanged. Nothing to do		alert box
I-WAN-05	dashboard view - tab "Administration - Settings"	administrator tries to update general settings without changing any items	no item selected	message	INFO: Systems unchanged. Nothing to do		alert box
I-WAN-06	dashboard view - tab "Administration - Settings"	administrator tries to update system settings	impossible to store operation to history	message	Impossible to store the operation in History		alert box

Table 4: Web Application Navigation info

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
W-WAN-01	signup	user try to register	unable to send confirmation email	message	WARNING! Your request has been successfully submitted but it is not possible to access to the mail server. Please, directly contact AIDA admin to check your request	Redirect to index	alert box
W-WAN-02	dashboard view - tab "Administration - Settings"	user try to Export data	export completed but missing stored items	message	WARNING! Backup file has been generated but the following items have been not exported: <list of missing items>		alert box

Table 5: Web Application Navigation warnings

13.2 Plots Generation

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-PG-01	trend / scatter / histogram / statistics / machine learning	user fills configuration form and submit	user does not fill a required field	error text appears	This field is required		form
E-PG-02	trend / scatter / histogram / statistics / machine learning	user fills configuration form and submit	user fills a field with a value having not allowed format	error text appears	This field must be <right format>		form
E-PG-03	trend / scatter / histogram / statistics / machine learning	user fills configuration form and submit	user fills a field with a value having not allowed format	error text appears	message with instructions on right format		form
E-PG-04	trend / scatter / histogram / statistics / machine learning	user fills configuration form and submit	AIDA configuration files are not in JSON format	message	ERROR! Unable to read configuration file. Please contact AIDA admin		alert box
E-PG-05	trend / scatter / histogram / statistics / machine learning	user fills configuration form and submit	AIDA configuration files are not in properly set	message	ERROR! Unable to read configuration file. Please contact AIDA admin		alert box
E-PG-06	trend / scatter / histogram / statistics / machine learning	user fills configuration form and submit	unable to connect to local DB	message	ERROR! Unable to connect to AIDA local database. Please retry or contact AIDA admin		alert box

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-PG-07	trend / scatter / histogram / statistics / machine learning	plot generation	unable to connect to remote DB	message	ERROR! Unable to connect to remote database. Please retry or contact AIDA admin		alert box
E-PG-08	trend / scatter / histogram / statistics / machine learning	plot generation offline	unable to store plot into local DB	mail	subject: ATTENTION: Failed <plot> generation. In text also reported : "Impossible to store data in local DB"		mail
E-PG-09	trend / scatter / histogram / statistics / machine learning	user tries to download csv file	unable to download	message	Error! Impossible to download file. Please, retry later or contact AIDA admin		alert box
E-PG-10	trend / scatter / histogram / statistics / machine learning	user tries to display image in Image explorer by using links in "Files" tab	unable to download image	message	ERROR! Impossible to download image from repository		alert box
E-PG-11	trend / scatter / histogram / statistics / machine learning	user fills configuration form and submit	user selects "Aggregated" both for CCD row and CCD col fields	error text appears	Aggregate option not allowed for both coord at the same time		form
E-PG-12	trend / scatter / histogram / statistics / machine learning	user fills configuration form and submit	wrong format for PID extra filter field	error text appears	This field must be a sequence of integer numbers separated by return		form

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-PG-13	trend / scatter / histogram / statistics / machine learning	user fills configuration form and submit	user sets End Date equal to Start Date	error text appears	End Date must be greater than Start Date		form
E-PG-14	trend / scatter / histogram / statistics / machine learning	user fills configuration form and submit	wrong format for "coeff" field	error text appears	Invalid format. Only single integer >0, comma separated integers >0 or range of integers >0 (a-b) are allowed		form
E-PG-15	trend / scatter / histogram / statistics / machine learning	plot generation	unable to get meta-data from remote DB	message	ERROR! Impossible to query remote archive to get metadata		alert box

Table 6: Plot Generation errors

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
I-PG-01	trend / scatter / histogram / statistics / machine learning	plot submission	date range is larger than defined settings	message	Data range is too large to compute online. An email will be send when analysis is completed		alert box
I-PG-02	trend / scatter / histogram / statistics / machine learning	plot generation	no data to plot	message	No available data for requested period		alert box

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
I-PG-03	trend / scatter / histogram / statistics / machine learning	plot generation	impossible to store operation to history	message	Impossible to store the operation in History		alert box
I-PG-04	view offline plot	page loading	user not logged on the web app	message	You must be logged to view plot. Please, login in a separate tab and then refresh this page.		text in related div
I-PG-04	view image analysis	page loading	user not logged on the web app	message	You must be logged to view results. Please, login in a separate tab and then refresh this page.		text in related div
I-PG-05	view image analysis	page loading	experiment expiration date has passed	message	No data to display. Expiration date has passed.		alert box

Table 7: Plot Generation info

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
W-PG-01	trend / scatter / histogram	user fills configuration form and submit	user does not choose any statistical function in "Advanced Statistics"	message	No statistical operation selected! Proceed anyway?		alert box
W-PG-02	trend / scatter / histogram / statistics / machine learning	plot generation	unable to connect to remote files archive by FTP	message	WARNING! Unable to download files by FTP. Analysis could be incomplete. Please, check connection parameters or contact AIDA admin.		alert box

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
W-PG-03	trend / scatter / histogram / statistics / machine learning	plot generation	unable to download remote files by FTP	message	WARNING! One or more files cannot be downloaded. Analysis could be incomplete		alert box
W-PG-04	trend / scatter / histogram / statistics / machine learning	plot generation	file cannot be opened to get data	message	WARNING! One or more files cannot be opened. Analysis could be incomplete		alert box

Table 8: Plot Generation warnings

13.3 Report Generation

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-RG-01	report generation by editor/update config file	configuration file validation	missing configuration filename	error text appears	This field is required.		form
E-RG-02	report generation by upload	upload file confirmation	uploaded file is not with allowed extension	message	Impossible to upload file! Extension not allowed: <filename> (<extension type>)		alert box
E-RG-03	report generation by upload	upload file confirmation	file not uploaded	message	Unable to upload the file. Try later. If the problem persists please contact AIDA admin.		alert box
E-RG-04	report generation by upload	upload file confirmation	configuration data not stored in local DB	message	Impossible to store configuration data in the local DB. Try later. If the problem persists please contact AIDA admin.		alert box

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-RG-05	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	config file not valid JSON	message	JSON Error! The configuration is not a valid JSON text		alert box
E-RG-06	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	config file not valid GENERAL INFO	message	Value Value Error! Error reading "General Info" data. Please check the syntax and required keys		alert box
E-RG-07	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	config file DATES for on-demand report are greater than today	message	Value Error! Start date and/or calculated stop date must be before today. Please check.		alert box
E-RG-08	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	config file contains empty branches	message	ERROR! Branch <hktm/-science if applicable> for system <system> is empty. Please, remove it from configuration file.		alert box

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-RG-09	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	config file not valid SYSTEMS	message	Key Error! <system_name> is not an allowed key		alert box
E-RG-10	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	config file not valid HKTM/-SCIENCE origin	message	Key Error! <origin_name> is not an allowed key for system <system_name>		alert box
E-RG-11	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	config file not valid detector (when applicable)	message	Key Error! Invalid syntax <detector> as Detector key in: <system>/<origin>		alert box
E-RG-12	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	config file not valid subsystem (when applicable)	message	Key Error! '<subsystem>' is not an allowed key for system ' <system>'		alert box

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-RG-13	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	missing required keys in experiment configuration (for example type of analysis)	message	Key Error "<key>" not found in <branch>		alert box
E-RG-14	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	missing required settings (for example, X key for scatter plot, Bin Size/Number of Bins for histogram)	message	Missing required parameters in <branch>		alert box
E-RG-15	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	required settings not allowed	message	Unrecognized filter "<wrong filter>" in <branch>		alert box
E-RG-16	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	missing required "Parameters" section for statistical functions	message	Missing required "Parameters" section in <branch>		alert box

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-RG-17	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	config file not valid parameter	message	Key Error! '<parameter>' is not an allowed key for <system>/<origin>		alert box
E-RG-18	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	config file not valid "Operation" key	message	Key Error! Invalid syntax for one or more "Operation" keys in <branch>		alert box
E-RG-19	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	config file not valid plot	message	Value Error! "Type" value not allowed in <branch>		alert box
E-RG-20	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	invalid format for required settings	message	Value Error! "<required>" is not in <right_format> <branch>		alert box

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-RG-21	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	invalid format for additional parameters list	message	Value Error! "Additional Parameters" value must be a list in <branch>		alert box
E-RG-22	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	invalid additional parameter	message	Value Error! One or more "Additional Parameters" in <branch> are not allowed		alert box
E-RG-23	report generation	launch/resume run report	unable to connect to local DB	message	ERROR! Impossible to start report generation. Impossible to connect to local DB. Please, contact AIDA admin		alert box
E-RG-24	report generation	report running	unable to connect to local DB	message	Impossible to connect to local DB		XML - PDF error list
E-RG-25	report generation	report running	unable to get meta-data from remote DB	message	Impossible to download file list from remote DB for dates in <date_range>		XML - PDF error list
E-RG-26	report generation	report running	xml not created	mail	subject: ATTENTION: Failed <period> report generation. In text also reported : "No XML report generated"		email

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-RG-27	report generation	report running	all analysis for a specific <system> cannot be performed during xml generation	message	Impossible to store results in XML file. Analysis cannot be performed.		XML - PDF error list
E-RG-28	report generation	report running	During xml generation, missing connection to local DB for all processes	message	Impossible to connect to retrieved data. Analysis can not be performed		XML - PDF error list
E-RG-29	kill/pause report	report stopping	unable to connect to local DB	message	Impossible to stop report <id>. ERROR : Impossible to connect to local DB. Please, contact AIDA admin		alert box
E-RG-30	kill/pause report	report stopping	unable to stop process	message	Impossible to stop report <id>. ERROR : Impossible to stop process with PID = <pid>. Please, contact AIDA admin		alert box
E-RG-31	re-enabling ingestion	administrator try to re-enable ingestion for a report	AIDA can not update local DB	message	Error! Impossible to re-enable to ingestion Report <report name>		alert box
E-RG-32	configuration file deletion	user try to delete report configuration file	unable to delete file	message	Error deleting file <filename> : <error> SQL: <query> PATH: <filepath>		alert box

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-RG-33	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	invalid format for Time Window	message	"Time Window" Type Error! Input value is not a number		alert box
E-RG-34	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	Invalid Sampling value	message	Value Error! Not allowed value for "Sampling"		alert box
E-RG-35	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	Invalid Start Time format	message	Date Format Error! Incorrect date format, should be yyyy-mm-ddThh:mm:ss		alert box
E-RG-36	report generation by upload/-configuration file creation by editor/update config file	configuration file validation	"Time Window" is less or equal to 0	message	"Time Window" Value Error! Input values is not greater than 0		alert box

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-RG-37	report generation by existing config file	configuration file validation	configuration file not created for current operating mode	message	Error! Impossible to use config file in current Operating Mode<current opmode>. Configuration file created for <config file opmode> Operating Mode		alert box
E-RG-38	report ingestion	user tries to ingest report to EAS	impossible to ingest	message	Error! Impossible to store report into remote archive. Please retry or contact IOT team.		alert box
E-RG-39	report ingestion	user tries to ingest report to EAS	impossible to create metadata xml	message	Error! Impossible to create metadata XML. Please contact IOT Team.		alert box
E-RG-40	report ingestion	user tries to ingest report to EAS	impossible to upload reports file	message	Impossible to ingest reports files. Please contact IOT Team		alert box
E-RG-41	report ingestion	user tries to ingest report to EAS	impossible to ingest metadata xml	message	ERROR! Impossible to ingest XML file		alert box

Table 9: Report Generation errors

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
W-RG-01	report generation by upload/-configuration file creation by editor	configuration file validation	uploaded filename already exists	message	A file with name '<filename>' already exists. The new file has been changed in <filename>_<current_datetime>		alert box

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
W-RG-02	report generation	report running	unable to download data files from remote DB (data from file branch)	message	Impossible to download one or more files from data DB. Analysis could be incomplete		XML - PDF error list
W-RG-03	report generation	report running	unable to read data from downloaded files (data from file branch)	message	Impossible to read one or more data files. Analysis could be incomplete		XML - PDF error list
W-RG-04	report generation	report running	unable to connect to remote data db (data from db branch)	message	Impossible to connect to remote DB for dates in <date range>		XML - PDF error list
W-RG-05	report generation	report running	unable to read data from remote DB (data from DB branch)	message	Impossible to read data from remote DB. Analysis could be incomplete.		XML - PDF error list
W-RG-06	report generation	report running	pdf not created	mail	PDF version is not available. Contact AIDA admin for further info.		email

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
W-RG-07	report generation	report running	some analysis for a specific <system> cannot be performed during xml generation	message	Impossible to store some results in XML file. Analysis could be incomplete.		XML - PDF error list
W-RG-08	report generation	report running	During xml generation, missing connection to local DB for some processes	message	Impossible to connect to some retrieved data. Analysis could be incomplete.		XML - PDF error list
W-RG-09	report generation	report running	no data for a key parameter	message	No data available for parameter <parameter> during acquisition #<acquisition>		XML - PDF error list
W-RG-10	report generation	report running	no data for X parameter for scatter plot	message	Parameter : <key parameter>. 'X' parameter has no data available for Acquisition #<acquisition>, Operation #<operation_id>		XML - PDF error list
W-RG-11	report generation	report running	a parameter has out-of-range values	message	<out-of-range number> values out of range for parameter <parameter> during acquisition #<acquisition>		XML - PDF error list

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
W-RG-12	report generation	report running	no data for an additional parameter	message	Parameter : <key parameter>. One or more additional parameters have no data available for Acquisition #<acquisition>, Operation #<operation_id>		XML - PDF error list
W-RG-13	report generation	report running	one of the foreseen plots for an operation cannot be produced	message	Impossible to generate all the foreseen <plot type plots> for Parameter <parameter>, Acquisition #<acquisition>, Operation #<operation_id>		XML - PDF error list
W-RG-14	report generation	report running	impossible to perform a statistical analysis	message	Impossible to perform Operation #<operation_id> for parameter <parameter>, Acquisition #<acquisition>		XML - PDF error list
W-RG-15	kill/pause report	report stopping	report stopped but some subprocesses still running	message	Report generation <id> stopped. WARNINGS : Subprocesses with pids = <pid> are still running. Please, contact AIDA admin		alert box
W-RG-16	kill/pause report	report stopping	report stopped but it's impossible to remove temporary data from DB	message	Report generation <id> stopped. WARNINGS : Impossible to remove temporary data from local DB. Please, contact AIDA admin		alert box

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
W-RG-17	kill/pause report	report stopping	report stopped but it's impossible to remove temporary directory	message	Report generation <id> stopped. WARNINGS : Impossible to remove report temporary directory. Please, contact AIDA admin		alert box
W-RG-18	kill/pause report	report stopping	report stopped but it's impossible change configuration file status from running to idle	message	Report generation <id> stopped. WARNINGS : Impossible to update config file status. Please, contact AIDA admin		alert box
W-RG-19	kill/pause report	report stopping	report stopped but it's impossible update running reports table	message	Report generation <id> stopped. WARNINGS : Impossible to update running reports status. Please, contact AIDA admin		alert box
W-RG-20	report ingestion	user tries to ingest report to EAS	report ingestion ok, but files could have not been uploaded	message	WARNING! Something has gone wrong during ingestion, but data could be successfully uploaded. Please check or contact IOT team.		alert box

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
W-RG-21	report ingestion	user tries to ingest report to EAS	report ingestion ok, but one or more files have not been uploaded	message	Warning! Report ingested but one or more files could not be uploaded		alert box

Table 10: Report Generation warnings

13.4 Flagging

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-FLA-001	image flagging	submitting image flag	flag not assigned	message	Impossible to store image into the AIDA archive. Please retry later or contact AIDA admin		alert box
E-FLA-002	report flagging	submitting report flag	flag not assigned	message	Impossible to flag report into the AIDA archive. Please retry later or contact AIDA admin.		alert box

Table 11: Flagging errors

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
I-FLA-001	image flagging	submitting image flag	impossible to send email to contact person	message	Image has been successfully stored into the AIDA archive but it is impossible to send an email to: <email>		alert box

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
I-FLA-002	image/report flagging	storing operation in history	Impossible to store the operation in History	message	Impossible to store the operation in History		alert box
I-FLA-003	report flagging	submitting report flag	impossible to send email to contact person	message	Report has been successfully stored into the AIDA archive but it is impossible to send an email to: <email>		alert box

Table 12: Flagging info

13.5 Analyze Local Data

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-ALD-01	file upload	user tries to upload not FITS or CSV file	file format not allowed	message	Error! Impossible to upload the file	page reloaded	alert box
E-ALD-02	file upload	user tries to upload a file with right format but indicating another file type	wrong file format	message	Error! Impossible to upload the file	page reloaded	alert box
E-ALD-03	file upload	user tries to upload file	impossible to server to load file	message	Error! Impossible to upload the file	page reloaded	alert box
E-ALD-04	file upload	user uploads empty file	empty file	message	Error! No data in uploaded file	page reloaded	alert box
E-ALD-05	analysis from local file	user fills configuration form and submit	user does not fill a required field	error text appears	This field is required		form

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-ALD-06	analysis from local file	user fills configuration form and submit	All data of a selected column are NULL or strings	message	No data to plot! Data are NULL or STRINGS		alert box

Table 13: Analyze Local Data errors

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
I-ALD-01	load local data	user does not select file to load	no file selected	message	Please, select a valid file		alert box

Table 14: Analyze Local Data info

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
W-ALD-01	file upload	user uploads file	impossible to store file info into local DB	message	Warning! File has been uploaded, but it is impossible to store it in the local temporary files DB.		alert box
W-ALD-02	file upload	user uploads an already existing file	file already uploaded	message	Warning! A file with name <filename> already exists in local temporary files. It has been overwritten.		alert box

Table 15: Analyze Local Data warnings

13.6 Web Application Installation

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-WAI-01	installation	user try to upload backup file	uploaded file is not a valid tar.gz	message	ERROR! Uploaded file is not a valid tar.gz		alert box
E-WAI-02	installation	user try to import data from uploaded backup file	impossible to read backup file or to import all the data	message	ERROR! Impossible to import data from backup file. Please, retry later or contact AIDA Admin.		alert box
E-WAI-03	installation	user try to import data from uploaded backup file	general communication error between frontend and backend	message	ERROR! Something has gone wrong. Impossible to finalize installation. Please, retry or contact AIDA staff.		alert box
E-WAI-07	installation	user submits Mail server configuration form	inserted data are not correct	message	ERROR! SMTP connection check failed. Please check your settings.		alert box
E-WAI-04	installation	user submits Mail server configuration form	smtp configuration file creation fails	message	Error storing SMTP data. Please retry or contact AIDA support team.		alert box
E-WAI-05	installation	user tries to abort installation	impossible to restore initial AIDA configuration	message	ERROR! Impossible to restore application. You can start a brand new installation to overwrite stored settings.		alert box
E-WAI-06	installation	user confirm account activation by email link	impossible to activate user	message	ERROR! Impossible to activate user. Please contact AIDA support team.		alert box

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
E-WAI-07	installation	user try to install web app	unable to connect to local DB	message	ERROR! Impossible to connect to local database. Installation aborted		alert box

Table 16: Web Application Installation errors

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
I-WAI-01	installation	user try to import data from uploaded backup file	user does not select any section	message	Nothing to import. Please, select at least one item to import.		alert box
I-WAI-02	installation	user try to import data from uploaded backup file	user does not select "Users" section to import	message*	<default message success/import> + INFO : No Users data imported. You will have to set the first Administrator in the following steps.		alert box
I-WAI-03	installation	user try to import data from uploaded backup file	user does not select "SMTP settings" section to import	message*	<default message success/import> + INFO : No SMTP server settings imported. You will have to set it in the following steps.		alert box
I-WAI-04	installation	administrator tries to install while at the same time another administrator has started installation procedure	installation already started	message	Installation process already completed while this windows was open. Redirecting...	Redirect to index	alert box

Table 17: Web Application Installation info.
*both in case of success and warning import

ID	Use case	Action catching error	Error	Handling action	Message	After message	Action location
W-WAI-01	installation	user try to import data from uploaded backup file	some data can not be imported	message	WARNING! Data import has been completed but the following items have been not imported: <list of not imported sections>		alert box
W-WAI-02	installation	user try to import data from uploaded backup file	"Users" section is not available in backup file	message	Missing Users data. Once import is completed, you should register the first admin. Proceed anyway?	wait for confirm	alert box
W-WAI-03	installation	user confirm account activation by email link	impossible to set notification email	message	WARNING! Impossible to set notification email. You can set it from admin dashboard successively.		alert box
W-WAI-04	installation	user sets info for first admin registration	impossible to set notification email	message	WARNING! Impossible to set notification email. You can set it from admin dashboard successively. Now you can close this page.		alert box

Table 18: Web Application Installation warnings

APPENDIX

A docker-compose

```
version: "2.4"
```

```
services:
```

```
  ioda:
```

```
    image: httpd:bullseye
```

```
    build: .
```

```
    ports:
```

```
      - "HostAidaPort:80"
```

```
    command: apache2ctl -D FOREGROUND
```

```
    volumes:
```

```
      - YourLocalAidaCodeDirectory:/var/www/html/aida
```

```
    links:
```

```
      - "db:mysql-aida"
```

```
    environment:
```

```
      UPLOAD_LIMIT: 10G
```

```
    cpus: numberOfCpusThatYouWantToDedicateToAida
```

```
  db:
```

```
    image: mysql:8.0.31
```

```
    hostname: mysql-aida
```

```
    command: --default-authentication-plugin=mysql_native_password
```

```
    --sql-mode=STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,  
    NO_ENGINE_SUBSTITUTION --max_connections=1000
```

```
    security_opt:
```

```
      - seccomp:unconfined
```

```
    restart: always
```

```
    volumes:
```

```
      - YourLocalAidaDbDirectory:/var/lib/mysql
```

```
    environment:
```

```
      - MYSQL_ALLOW_EMPTY_PASSWORD=1
```

```
    ports:
```

```
      - "HostDBPort:3306"
```

```
    cpus: 1
```

dbfake:

image: mysql:8.0.31

hostname: aida-fake

command: --default-authentication-plugin=mysql_native_password --sql-mode=STRICT_TRANS_TABLES

security_opt:

- seccomp:unconfined

environment:

- MYSQL_ALLOW_EMPTY_PASSWORD=1

restart: always

volumes:

- YourLocalFakeDataDbDirectory:/var/lib/mysql

ports:

- "3307:3306"

expose:

- 3307

cpus: 1

B Dockerfile

```
RUN apt-get update && apt-get install -y \  
php \  
libapache2-mod-php \  
php-mysql \  
php7.4-gd \  
python3 \  
python3-pip \  
python-is-python3 \  
saods9 \  
xvfb
```

```
RUN a2enmod cgid
```

```
RUN phpenmod mysql
```

```
RUN pip3 install \  
astropy==4.2.1 \  
certifi==2021.10.8 \  
cffi==1.14.5 \  
charset-normalizer==2.0.7 \  
cryptography==3.4.7 \  
cssselect2==0.4.1 \  
cycller==0.10.0 \  
DBUtils==2.0.1 \  
h5py==3.2.1 \  
idna==3.3 \  
#joblib==1.0.1 \  
joblib==1.5.1 \  
kiwisolver==1.3.1 \  
lxml==4.6.3 \  
matplotlib==3.4.1 \  
memory-profiler==0.58.0 \  
numpy==1.22.4 \  
Pillow==8.2.0 \  

```

```
psrecord==1.2 \  
psutil==5.8.0 \  
pyparser==2.20 \  
pyerfa==1.7.2 \  
PyMySQL==1.0.2 \  
pyparsing==2.4.7 \  
pypdf2==3.0.1 \  
python-dateutil==2.8.1 \  
reportlab==3.5.67 \  
requests==2.26.0 \  
#scikit-learn==0.24.1 \  
scikit-learn==1.6.1 \  
scipy==1.6.2 \  
six==1.15.0 \  
sklearn==0.0 \  
svglib==1.1.0 \  
#threadpoolctl==2.1.0 \  
threadpoolctl==3.6.0 \  
tinycss2==1.1.0 \  
urllib3==1.26.7 \  
webencodings==0.5.1 \  
#pandas \  
lsst_efd_client
```

```
COPY apache2.conf /etc/apache2/apache2.conf  
COPY php.ini /etc/php/7.4/apache2/php.ini  
RUN export DISPLAY=:1  
RUN Xvfb :1 -screen 0 1024x768x16 &
```

```
EXPOSE 80  
EXPOSE 3306
```

```
CMD apache2ctl -D FOREGROUND
```


C Troubleshooting

- If AIDA is installed in a Virtual Machine the URL sent from AIDA will be meaningless if used outside from the VM itself;
- If AIDA is installed behind a firewall that blocks its port the URL sent from AIDA could be accessed only from machine behind the firewall;
- We noticed that if Podman is used in the place of Docker it could happen that the alias of the machine could be wrong, in this case you should replace the name of the db in the `config.json` file in the main root of AIDA:

```
"host" : "mysql-aida",
```

should become:

```
"host" : "db",
```

- AIDA needs to open some pop-up windows during data exploration. Enable pop-up windows for AIDA website in your browser.
- After the update to a new version, it could be necessary to refresh the browser cache to make the changes effective.

END OF DOCUMENT